Quantum Phase Estimation

A Complete and Comprehensive Lecture

Course: Quantum Computing
Level: Graduate / Advanced Undergraduate

Prerequisites: Linear algebra, Dirac notation, basic quantum gates, quantum Fourier transform at a conceptual level.

1. The *eigenvalue / phase problem* and why it matters.
2. How controlled-unitary gates encode phase information via *phase kickback*.
3. The full derivation of the QPE circuit from first principles, with motivation at every step.
4. The Quantum Fourier Transform (QFT) as a phase decoder and why its inverse is used.
5. Precision, success probability, and a rigorous error analysis.
6. A fully-traced numerical example: the T gate, step by step, with every coefficient written out.
7. What happens when the phase is *not* exactly representable.
8. Real-world applications: Shor's algorithm, quantum simulation, HHL, quantum counting.
9. Practical considerations: iterative QPE, noise, qubit count trade-offs.

Contents

1 Motivation: Why Eigenvalues Matter in Quantum Computing

1.1 The Central Problem

Many of the most important problems in science and engineering reduce to *eigenvalue problems*. In chemistry, the ground-state energy of a molecule is the smallest eigenvalue of its Hamiltonian. In number theory, the period of a function (the key to factoring large integers) is related to the eigenvalues of a unitary operator. In linear systems, stability is governed by eigenvalues.

Classically, finding eigenvalues of an $N \times N$ matrix costs at least $\mathcal{O}(N^2)$ time and often much more. For quantum systems, N grows *exponentially* in the number of particles — making the classical approach completely infeasible even for molecules with a few dozen electrons.

The key insight. Quantum Phase Estimation (QPE) solves a specific version of the eigenvalue problem: given a unitary operator U and (approximately) one of its eigenstates $|\psi\rangle$, estimate the corresponding eigenvalue $e^{2\pi i\varphi}$ to n bits of precision using only $\mathcal{O}(n)$ queries to a controlled-U oracle. This is exponentially faster than any known classical algorithm for the same task.

1.2 The Eigenvalue of a Unitary Operator

Recall from linear algebra:

Definition 1.1 ▶ Unitary Operator and Its Eigenvalues

An operator U on $\mathbb{C}^N$ is **unitary** if $U^\dagger U = UU^\dagger = \mathbb{1}$, where $U^\dagger$ denotes the conjugate transpose of U and $\mathbb{1}$ is the $N \times N$ identity operator. A non-zero vector $|\psi\rangle \in \mathbb{C}^N$ is an **eigenstate** of U with **eigenvalue** $\lambda \in \mathbb{C}$ if

$$U|\psi\rangle = \lambda|\psi\rangle .$$

Because U is unitary, every eigenvalue satisfies $|\lambda| = 1$ (proved below), so we may write

$$\lambda = e^{2\pi i\varphi}, \quad \varphi \in [0, 1).$$

The real number φ is called the **(relative) phase** of the eigenvalue. It is defined modulo 1, i.e. φ and $\varphi + 1$ encode the same eigenvalue.

Why $|\lambda| = 1$? If $U|\psi\rangle = \lambda|\psi\rangle$, then applying the bra from the left and using unitarity:

$$\langle\psi| U^\dagger U |\psi\rangle = \langle\psi| |\psi\rangle = 1,$$

but also

$$\langle\psi| U^\dagger U |\psi\rangle = (U|\psi\rangle)^\dagger(U|\psi\rangle) = (\lambda|\psi\rangle)^\dagger(\lambda|\psi\rangle) = \lambda^*\lambda \langle\psi|\psi\rangle = |\lambda|^2.$$

Hence $|\lambda|^2 = 1$, so $|\lambda| = 1$. Every eigenvalue of every unitary operator lies on the complex unit circle, fully characterised by the single real number $\varphi \in [0, 1)$.

Rephrasing the goal. QPE finds φ given U and $|\psi\rangle$. Every quantum algorithm that claims an exponential speedup via QPE ultimately reduces its hard problem to: *"find this phase."* The rest of these notes explain, from first principles, exactly how.

1.3 Running Example Setup

Throughout this lecture we use the following concrete example, small enough to trace entirely by hand yet rich enough to illustrate every subtlety.

Running Example: the T Gate

The single-qubit T **gate** (also called the $\pi/8$ gate) is:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\pi/4} \end{pmatrix}.$$

Eigenvalues and eigenvectors of T: Since T is diagonal, its eigenvalues are its diagonal entries: $\lambda_0 = 1$ (eigenvector $|0\rangle$) and $\lambda_1 = e^{\mathrm{i}\pi/4}$ (eigenvector $|1\rangle$).

Identifying the phase φ: Writing $\lambda_1 = e^{2\pi\mathrm{i}\varphi}$:

$$e^{2\pi\mathrm{i}\varphi} = e^{\mathrm{i}\pi/4} \implies 2\pi\varphi = \frac{\pi}{4} \implies \varphi = \frac{1}{8}.$$

In binary: $\varphi = 1/8 = 0.001_2$ (since $\frac{0}{2} + \frac{0}{4} + \frac{1}{8} = \frac{1}{8}$).

QPE setup: We run QPE with $U = T$, eigenstate $|\psi\rangle = |1\rangle$, and $n = 3$ ancilla qubits. With $n = 3$, we can represent $2^3 = 8$ distinct values $\{0, \frac{1}{8}, \frac{2}{8}, \dots, \frac{7}{8}\}$. Since $\varphi = \frac{1}{8}$ is exactly representable, we expect to measure the binary string 001 (encoding the integer $f = 1$, giving $\tilde{\varphi} = 1/8$) *with probability 1*.

2 Prerequisites and Notation

Even though students in this course know the basics, it is worth fixing notation precisely — small sign or convention errors accumulate and produce incorrect final results.

2.1 The Computational Basis and Dirac Notation

An n-qubit **computational basis state** is written $|j\rangle$ where $j \in \{0, 1, \ldots, 2^n - 1\}$ is the decimal value of the corresponding binary string. The binary string is ordered from *most significant bit (MSB) on the left* to *least significant bit (LSB) on the right*:

$$|j\rangle = |j_{n-1}\, j_{n-2} \cdots j_1\, j_0\rangle\,, \quad j = \sum_{k=0}^{n-1} j_k\, 2^k, \quad j_k \in \{0, 1\}.$$

For example, with $n = 3$: $|6\rangle = |110\rangle$ since $1 \cdot 4 + 1 \cdot 2 + 0 \cdot 1 = 6$.

2.2 Binary Fractions

A crucial notational convenience: the **binary fraction**

$$0.b_1 b_2 \cdots b_n \;:=\; \sum_{k=1}^{n} \frac{b_k}{2^k} = \frac{b_1}{2} + \frac{b_2}{4} + \cdots + \frac{b_n}{2^n}, \quad b_k \in \{0, 1\}.$$

Examples: $0.1_2 = 1/2$; $0.01_2 = 1/4$; $0.101_2 = 1/2 + 1/8 = 5/8$.

The phase φ will often be expressed as such a fraction. Note the relationship to the integer encoding: if the phase register contains the integer $j = j_{n-1} \cdots j_0$ in binary, then

$$\frac{j}{2^n} = 0.j_{n-1}\, j_{n-2} \cdots j_0.$$

2.3 The Quantum Fourier Transform (QFT)

QPE uses the QFT as a subroutine. We review it carefully here.

Definition 2.1 ▶ Quantum Fourier Transform

The n-qubit **Quantum Fourier Transform** (QFT) is the unitary operator defined by its action on computational basis states:

$$\mathrm{QFT}\,|j\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi \mathrm{i} jk/2^n}\,|k\rangle\,, \qquad j \in \{0, \ldots, 2^n - 1\}.$$

The **inverse QFT** (QFT^{-1}) satisfies $\mathrm{QFT}^{-1} = \mathrm{QFT}^\dagger$ and acts as:

$$\mathrm{QFT}^{-1}\,|j\rangle = \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{-2\pi \mathrm{i} jk/2^n}\,|k\rangle\,.$$

In the **product form** (essential for circuit implementation), the QFT of a computational basis state $|j_1 j_2 \cdots j_n\rangle$ (where j_1 is the MSB and j_n is the LSB) is:

$$\mathrm{QFT}\,|j_1 j_2 \cdots j_n\rangle = \frac{1}{\sqrt{2^n}} \Big(|0\rangle + e^{2\pi \mathrm{i}\, 0.j_n}\,|1\rangle\Big) \otimes \Big(|0\rangle + e^{2\pi \mathrm{i}\, 0.j_{n-1} j_n}\,|1\rangle\Big) \otimes \cdots \otimes \Big(|0\rangle + e^{2\pi \mathrm{i}\, 0.j_1 j_2 \cdots j_n}\,|1\rangle\Big).$$

Why the product form matters. The product form shows that each output qubit of the QFT carries phase information for exactly one bit of the binary fraction $0.j_1 j_2 \cdots j_n$. The *last* output qubit ($k = n$) encodes only j_n (one bit), while the *first* output qubit ($k = 1$) encodes all n bits. This structure is what makes the QFT efficient to implement (in $\mathcal{O}(n^2)$ gates) and is also what allows QPE to decode the phase: Stage 2 of QPE prepares a state that is *exactly* in this product form, and the QFT^{-1} then reads it out as a basis state.

2.4 Controlled-Unitary Gates

Definition 2.2 ▶ Controlled-U Gate

Let U be a unitary acting on m qubits. The **controlled-U** gate c-U acts on one control qubit and m target qubits as:

$$c\text{-}U \ |c\rangle\,|x\rangle = \begin{cases} |0\rangle\,|x\rangle & \text{if } c = 0, \\ |1\rangle\,U\,|x\rangle & \text{if } c = 1. \end{cases}$$

In operator form: $c\text{-}U = |0\rangle\langle 0| \otimes \mathbb{1}_m + |1\rangle\langle 1| \otimes U$. Here $\mathbb{1}_m$ is the identity on the m-qubit target space.

2.4.1 Phase Kickback. Now observe the crucial phenomenon that underlies all of QPE. Let the control qubit be in the equal superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and let the target be in the eigenstate $|\psi\rangle$ with $U\,|\psi\rangle = e^{2\pi i\varphi}\,|\psi\rangle$:

$$\begin{aligned} c\text{-}U\ \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |\psi\rangle &= \frac{|0\rangle \otimes \mathbb{1}\,|\psi\rangle + |1\rangle \otimes U\,|\psi\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle\,|\psi\rangle + e^{2\pi i\varphi}\,|1\rangle\,|\psi\rangle}{\sqrt{2}} \\ &= \frac{|0\rangle + e^{2\pi i\varphi}\,|1\rangle}{\sqrt{2}} \otimes |\psi\rangle\,. \end{aligned} \tag{1}$$

Phase Kickback. Equation (1) is the single most important mechanism in QPE. The phase $e^{2\pi i\varphi}$ *migrates from the target to the control qubit*, while the target $|\psi\rangle$ is returned *completely unchanged.*
This is why:

- We use $|\psi\rangle$ as an eigenstate (not a general state): kickback only produces a clean phase factor when the target is an eigenstate, because $U\,|\psi\rangle = e^{2\pi i\varphi}\,|\psi\rangle$ leaves $|\psi\rangle$ undisturbed.
- The eigenstate register remains reusable throughout all controlled-U^{2^k} applications.

- Information about φ is deposited into the *phase register* (control qubits), where it can be extracted by the QFT^{-1}.

3 The QPE Circuit: Structure and Deep Intuition

3.1 Circuit Overview

QPE uses two registers:

- **Phase register** (ancilla): n qubits, initialised to $|0\rangle^{\otimes n}$. After the algorithm, measuring this register yields an n-bit binary string encoding $\tilde{\varphi} \approx \varphi$.
- **Eigenstate register**: m qubits holding $|\psi\rangle$, the eigenstate of U. This register is untouched by the measurement.

The circuit has three stages:

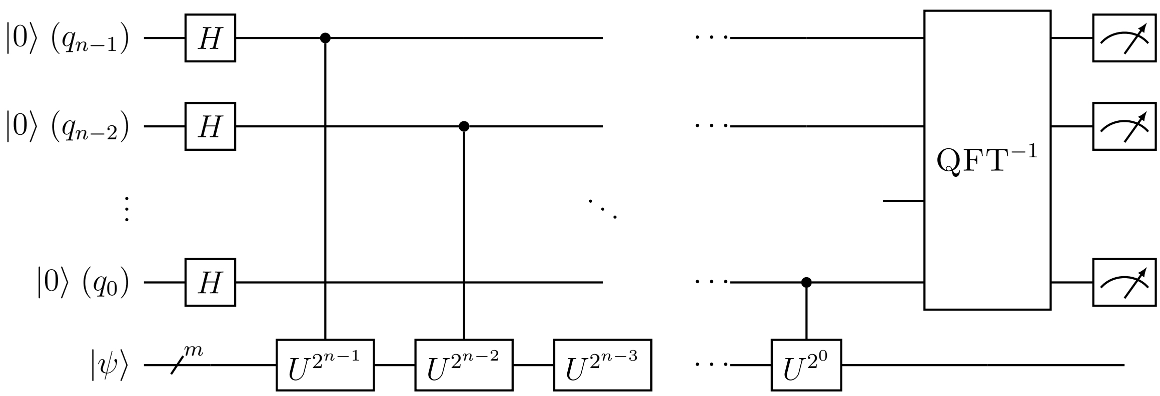
Stage 1. Superposition: Apply $H^{\otimes n}$ to the phase register. *Why:* to create a uniform superposition that will allow all n phase-kickback operations to work simultaneously.

Stage 2. Controlled rotations: Apply c-U^{2^k} for $k = 0, 1, \ldots, n-1$. *Why:* to encode the phase φ into the relative amplitudes of the phase register, in exactly the form produced by the QFT.

Stage 3. Inverse QFT: Apply QFT^{-1} to the phase register, then measure. *Why:* to convert the QFT-encoded state back into a computational basis state that directly reads out φ.

3.2 The Circuit Diagram

Below, qubit q_{n-1} is the MSB of the phase register (top wire) and q_0 is the LSB (bottom ancilla wire). The eigenstate register occupies the bottom m wires.



Reading the diagram:

- The top ancilla qubit (q_{n-1}, MSB) controls $U^{2^{n-1}}$, the *highest* power.
- The bottom ancilla qubit (q_0, LSB) controls $U^{2^0} = U$, the *lowest* power.
- After all controlled gates, the n ancilla qubits undergo the QFT^{-1}, and each is then measured in the computational basis.
- The eigenstate register $|\psi\rangle$ exits unchanged.

3.3 Stage 1 — Creating Superposition: Why Hadamard?

After applying $H^{\otimes n}$ to $|0\rangle^{\otimes n}$, the phase register becomes

$$H^{\otimes n}|0\rangle^{\otimes n} = \bigotimes_{k=0}^{n-1} H|0\rangle = \bigotimes_{k=0}^{n-1} \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} |j\rangle .$$

The last equality follows because expanding n factors of $(|0\rangle + |1\rangle)$ produces all 2^n binary strings, each with coefficient $1/\sqrt{2^n}$.

Why superposition is essential. If the phase register started in $|0\rangle^{\otimes n}$ (no Hadamard), the controlled-U^{2^k} gates would do nothing: the control qubit is $|0\rangle$, so U is never applied, and no phase information can be extracted.
The Hadamard puts each control qubit into the superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Now the $|1\rangle$ branch of each qubit *activates* the controlled gate, while the $|0\rangle$ branch does not. Phase kickback then places the phase $e^{2\pi i\varphi 2^k}$ as a relative phase on the $|1\rangle$ branch. This is the only way to encode continuous phase information into a discrete quantum state.

3.4 Stage 2 — Controlled-U^{2^k} Gates: Why These Powers?

Let qubit k (the k-th ancilla from the LSB, $k = 0, 1, \ldots, n-1$) control the gate U^{2^k}. Before Stage 2, the full system state is

$$\frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} |j\rangle \otimes |\psi\rangle .$$

By phase kickback (Eq. (1)), applying controlled-U^{2^k} transforms qubit k as follows (since $U^{2^k}|\psi\rangle = e^{2\pi i\varphi \cdot 2^k}|\psi\rangle$):

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \longrightarrow \frac{|0\rangle + e^{2\pi i\varphi \cdot 2^k}|1\rangle}{\sqrt{2}},$$

while $|\psi\rangle$ remains unchanged. After all n controlled gates, the phase register is in the product state:

$$\bigotimes_{k=0}^{n-1} \frac{|0\rangle + e^{2\pi i\varphi 2^k}|1\rangle}{\sqrt{2}}.$$

Expanding this tensor product and using $j = \sum_{k=0}^{n-1} 2^k j_k$:

$$
\begin{aligned}
|\Psi_2\rangle &= \bigotimes_{k=0}^{n-1} \frac{|0\rangle + e^{2\pi \mathrm{i}\varphi 2^k}|1\rangle}{\sqrt{2}} \otimes |\psi\rangle \\
&= \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} \prod_{k=0}^{n-1} \left(e^{2\pi \mathrm{i}\varphi 2^k}\right)^{j_k} |j\rangle \otimes |\psi\rangle \\
&= \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i}\varphi \sum_{k=0}^{n-1} 2^k j_k} |j\rangle \otimes |\psi\rangle \\
&= \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i}\varphi j} |j\rangle \otimes |\psi\rangle . \qquad (2)
\end{aligned}
$$

Why powers of 2? The crucial step is the last line of the derivation above: the exponent simplifies to φj only because $j = \sum_k 2^k j_k$ is the binary expansion of j. By choosing the k-th gate to apply U^{2^k}, we ensure that the accumulated phase for basis state $|j\rangle$ is exactly $e^{2\pi \mathrm{i}\varphi j}$ — precisely the form required by the QFT. Any other choice of exponents would break this identity and the QFT^{-1} would not produce a sharp peak.

3.5 Stage 3 — Inverse QFT: Why It Decodes the Phase

After Stage 2, the phase register holds (ignoring $|\psi\rangle$):

$$|\Phi\rangle = \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i}\varphi j} |j\rangle .$$

Now compare this with the definition of the QFT:

$$\mathrm{QFT}\,|f\rangle = \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i} f j/2^n} |j\rangle .$$

Setting $f = \varphi \cdot 2^n$, we see that $|\Phi\rangle = \mathrm{QFT}\,|f\rangle$. Therefore:

$$\mathrm{QFT}^{-1}|\Phi\rangle = \mathrm{QFT}^{-1}\big(\mathrm{QFT}\,|f\rangle\big) = |f\rangle .$$

The QFT^{-1} as a decoder. Stage 2 prepares a state that is *exactly the QFT of the basis state* $|f\rangle$ where $f = \varphi \cdot 2^n$. Applying the QFT^{-1} simply undoes this transform, recovering $|f\rangle$. Measuring the phase register then gives f, and $\tilde{\varphi} = f/2^n \approx \varphi$. This is the quantum analogue of the inverse FFT: Stage 2 prepares a frequency-domain signal (a "spike at frequency φ"), and the QFT^{-1} converts it back to the time domain, yielding the basis state $|f\rangle$.

Why QFT^{-1} and not QFT? Because Stage 2 prepares $\mathrm{QFT}\,|f\rangle$, not $|f\rangle$ itself. We want to go backward through the QFT, recovering $|f\rangle$ from its QFT image. That backward step is the *inverse* QFT. Applying the forward QFT again would produce $\mathrm{QFT}(\mathrm{QFT}\,|f\rangle)$, which is a (bit-reversed) permutation of $|f\rangle$ — not the sharp recovery we want.

This exact analysis requires $\varphi = f/2^n$ for an integer f, i.e. φ is an *exact n-bit binary fraction.* When this condition holds, QPE succeeds with probability 1. When it does not, we obtain a probability distribution peaked near $\lfloor \varphi \cdot 2^n \rceil$ (the nearest integer), and the success probability has a lower bound of $4/\pi^2 \approx 40.5\%$. Section 5 analyses this in full detail.

4 Complete Step-by-Step Circuit Analysis

We now trace the circuit *algebraically*, step by step, for a general n-qubit phase register.

4.1 Initial State

$$|\Psi_0\rangle = \underbrace{|0\rangle^{\otimes n}}_{\text{phase register}} \otimes \underbrace{|\psi\rangle}_{\text{eigenstate}} .$$

4.2 After the Hadamard Layer

Apply H to each of the n phase qubits independently:

$$|\Psi_1\rangle = H^{\otimes n} |0\rangle^{\otimes n} \otimes |\psi\rangle = \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} |j\rangle \otimes |\psi\rangle .$$

4.3 After the Controlled-U^{2^k} Gates

Proposition 4.1

After applying controlled-U^{2^k} (control: ancilla qubit k, target: eigenstate register) for $k = 0, 1, \ldots, n-1$, the state is:

$$|\Psi_2\rangle = \frac{1}{\sqrt{2^n}} \sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i} \varphi j} |j\rangle \otimes |\psi\rangle .$$

Proof. Write the phase register in the tensor product basis as $|j\rangle = |j_{n-1}\rangle \otimes \cdots \otimes |j_0\rangle$. Qubit k is in the state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Applying controlled-$U^{2^k}$ to qubit k and the eigenstate register:

$$\frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |\psi\rangle \longmapsto \frac{|0\rangle \otimes |\psi\rangle + |1\rangle \otimes U^{2^k} |\psi\rangle}{\sqrt{2}} = \frac{|0\rangle + e^{2\pi \mathrm{i} \varphi 2^k} |1\rangle}{\sqrt{2}} \otimes |\psi\rangle .$$

The second equality uses $U^{2^k} |\psi\rangle = (e^{2\pi \mathrm{i}\varphi})^{2^k} |\psi\rangle = e^{2\pi \mathrm{i}\varphi 2^k} |\psi\rangle$, which holds because $|\psi\rangle$ is an eigenstate of U with eigenvalue $e^{2\pi \mathrm{i}\varphi}$. Different controlled gates act on different (non-overlapping) control qubits but the same eigenstate register; since each leaves $|\psi\rangle$ invariant, they act independently, and the phase register after all n gates is:

$$|\Psi_2\rangle = \bigotimes_{k=0}^{n-1} \frac{|0\rangle + e^{2\pi \mathrm{i}\varphi 2^k} |1\rangle}{\sqrt{2}} \otimes |\psi\rangle .$$

Expanding the tensor product and noting that the coefficient of $|j\rangle = |j_{n-1} \cdots j_0\rangle$ is

$$\frac{1}{\sqrt{2^n}} \prod_{k=0}^{n-1} \left(e^{2\pi \mathrm{i}\varphi 2^k}\right)^{j_k} = \frac{1}{\sqrt{2^n}} e^{2\pi \mathrm{i}\varphi \sum_k 2^k j_k} = \frac{1}{\sqrt{2^n}} e^{2\pi \mathrm{i}\varphi j},$$

the result follows. $\square$

4.4 After the Inverse QFT

We apply QFT^{-1} to the phase register. By definition:

$$\mathrm{QFT}^{-1}\,|j\rangle = \frac{1}{\sqrt{2^n}}\sum_{k=0}^{2^n-1} e^{-2\pi \mathrm{i} jk/2^n}\,|k\rangle\,.$$

By linearity, applying QFT^{-1} to $|\Psi_2\rangle$:

$$\begin{aligned}
|\Psi_3\rangle &= \mathrm{QFT}^{-1}\left[\frac{1}{\sqrt{2^n}}\sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i}\varphi j}\,|j\rangle\right]\otimes|\psi\rangle \\
&= \frac{1}{\sqrt{2^n}}\sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i}\varphi j}\left[\frac{1}{\sqrt{2^n}}\sum_{k=0}^{2^n-1} e^{-2\pi \mathrm{i} jk/2^n}\,|k\rangle\right]\otimes|\psi\rangle \\
&= \frac{1}{2^n}\sum_{k=0}^{2^n-1}\left[\sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i} j(\varphi - k/2^n)}\right]|k\rangle\otimes|\psi\rangle\,. \qquad (3)
\end{aligned}$$

Definition 4.1 ▶ Probability Amplitude and Measurement Probability

From Eq. (3), the **probability amplitude** of measuring outcome $k \in \{0,\ldots,2^n-1\}$ in the phase register is:

$$\alpha_k = \frac{1}{2^n}\sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i} j(\varphi - k/2^n)},$$

and the corresponding **measurement probability** is $P(k) = |\alpha_k|^2$. Note that $\sum_{k=0}^{2^n-1} P(k) = 1$ (probability normalisation), which follows from the unitarity of QFT^{-1}.

4.4.1 Case 1: φ is an exact n-bit binary fraction.

Suppose $\varphi = f/2^n$ for some integer $f \in \{0,\ldots,2^n-1\}$. Then:

$$\alpha_k = \frac{1}{2^n}\sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i} j(f-k)/2^n}.$$

This is a geometric series with common ratio $r = e^{2\pi \mathrm{i}(f-k)/2^n}$:

$$\sum_{j=0}^{2^n-1} r^j = \begin{cases} 2^n & r = 1,\ \text{i.e., } f = k \pmod{2^n}, \\ \dfrac{1-r^{2^n}}{1-r} = \dfrac{1-e^{2\pi \mathrm{i}(f-k)}}{1-e^{2\pi \mathrm{i}(f-k)/2^n}} = 0 & r \neq 1. \end{cases}$$

The numerator vanishes for $r \neq 1$ because $f-k$ is a non-zero integer, making $e^{2\pi \mathrm{i}(f-k)} = 1$. Therefore $\alpha_k = \delta_{kf}$, and:

$$\boxed{P(k) = \begin{cases} 1 & k = f, \\ 0 & k \neq f. \end{cases}}$$

Measuring gives f with certainty, and we recover $\varphi = f/2^n$ exactly.

4.4.2 Case 2: φ is not exactly representable.

Let $b = \lfloor \varphi \cdot 2^n \rceil$ be the nearest integer to $\varphi \cdot 2^n$ (ties broken arbitrarily). Write the deviation $\delta = \varphi - b/2^n \in (-1/2^{n+1}, 1/2^{n+1}]$. Then, using the geometric series formula and the identity $|1 - e^{i\theta}| = 2|\sin(\theta/2)|$:

$$P(b) = \frac{1}{4^n} \left| \sum_{j=0}^{2^n-1} e^{2\pi \mathrm{i} j\delta} \right|^2 = \frac{1}{4^n} \cdot \frac{\sin^2(\pi 2^n \delta)}{\sin^2(\pi\delta)}.$$

For $\delta = 0$ (exact case) this equals 1. To find the worst-case lower bound, we maximise $|\delta|$ subject to $|\delta| \leq 1/2^{n+1}$:

$$P(b) \;\geq\; \frac{1}{4^n} \cdot \frac{1}{\sin^2(\pi/2^{n+1})} \cdot \sin^2\!\left(\frac{\pi \cdot 2^n}{2^{n+1}}\right) = \frac{\sin^2(\pi/2)}{\big(2^n \sin(\pi/2^{n+1})\big)^2} = \frac{1}{\big(2^n \sin(\pi/2^{n+1})\big)^2}.$$

Using $\sin x \leq x$ gives $2^n \sin(\pi/2^{n+1}) \leq 2^n \cdot \pi/2^{n+1} = \pi/2$, so:

$$P(b) \;\geq\; \frac{1}{(\pi/2)^2} = \frac{4}{\pi^2} \approx 0.405.$$

Even in the worst case (phase exactly halfway between two representable values), QPE still succeeds with probability at least 40.5%. Repeating the algorithm r times and taking the median (or majority vote) increases the success probability to $1 - \epsilon$ for any $\epsilon > 0$, at a cost of $\mathcal{O}(\log(1/\epsilon))$ repetitions (by a Chernoff bound).

5 Precision, Success Probability, and Error Analysis

5.1 Precision of n Ancilla Qubits

With n ancilla qubits, the phase register can represent 2^n distinct values: $\{0, 1/2^n, 2/2^n, \ldots, (2^n - 1)/2^n\}$. The *resolution* (smallest gap between representable values) is $1/2^n$.

Theorem 5.1 ▶ QPE Precision Guarantee

Let U be a unitary with eigenstate $|\psi\rangle$ and eigenphase $\varphi \in [0, 1)$. Let $\epsilon, \epsilon' \in (0, 1)$ be desired parameters. With

$$n = \left\lceil \log_2\left(\frac{2}{\epsilon}\right) \right\rceil + \left\lceil \log_2\left(\frac{1}{\epsilon'}\right) \right\rceil \quad \text{ancilla qubits,}$$

the QPE circuit outputs an estimate $\tilde{\varphi} = k/2^n$ satisfying

$$|\tilde{\varphi} - \varphi| \le \epsilon'$$

with probability at least $1 - \epsilon$.

Simple rule of thumb: to estimate φ to t bits of precision with success probability at least $1 - 2^{-(s)}$ (for some safety margin $s \ge 1$), use $n = t + s$ ancilla qubits. Each additional qubit beyond t exponentially suppresses the failure probability.

5.2 The Probability Distribution

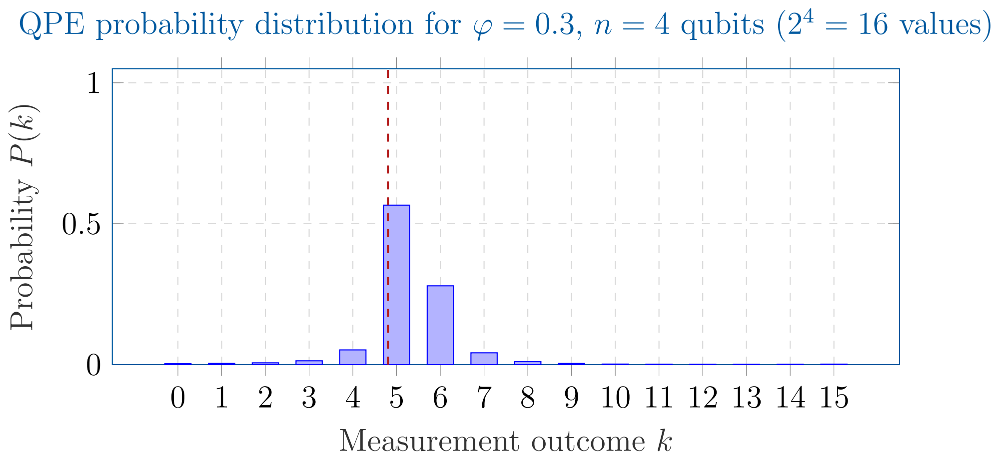


Figure 1: Probability distribution over measurement outcomes for $\varphi = 0.3$ and $n = 4$ qubits. Since $\varphi \cdot 16 = 4.8$, the nearest representable values are $k = 5$ ($\tilde{\varphi} = 5/16 = 0.3125$, error 0.0125) and $k = 4$ ($\tilde{\varphi} = 4/16 = 0.25$, error 0.05). The dominant peak at $k = 5$ has $P \approx 56.6\% \ > \ 4/\pi^2 \approx 40.5\%$, consistent with the worst-case lower bound. The second peak at $k = 4$ has $P \approx 5.2\%$. All probabilities sum to 1.

Table 1: QPE gate complexity with n ancilla qubits, where the primitive controlled-U can be implemented in g elementary gates.

Component	**Gate count**	**Notes**
Hadamard layer	n	One H per ancilla qubit
Controlled-U^{2^k} gates	$\sum_{k=0}^{n-1} g \cdot 2^k = g(2^n - 1)$	If implemented naively by repetition
Controlled-U^{2^k} gates	$\mathcal{O}(n \cdot g)$	If U^{2^k} is computed by repeated squaring
Inverse QFT	$\mathcal{O}(n^2)$ exact; $\mathcal{O}(n \log n)$ approximate	
Measurement	n	
Total	$\mathcal{O}(n^2 + ng)$	Using repeated squaring

5.3 Circuit Complexity

The power of repeated squaring. Computing U^{2^k} from U takes exactly k squarings: $U^2 = U \cdot U$, $U^4 = U^2 \cdot U^2$, etc. Thus $U^{2^{n-1}}$ costs only $n-1$ multiplications, not 2^{n-1} applications of U. This is why QPE costs $\mathcal{O}(n)$ oracle queries rather than $\mathcal{O}(2^n)$, yielding an exponential improvement in the number of calls to U.

6 Fully Worked Numerical Example: The T Gate ($n = 3$ Qubits)

We now execute the *complete* QPE circuit for the running example from Section 1.3, tracking every quantum state explicitly.

Setup Recap

- Unitary: $U = T = \begin{pmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\pi/4} \end{pmatrix}$
- Eigenstate: $|\psi\rangle = |1\rangle$
- Eigenvalue: $T\,|1\rangle = e^{\mathrm{i}\pi/4}\,|1\rangle$, so $\lambda = e^{2\pi\mathrm{i}/8}$, hence $\varphi = 1/8$
- Number of ancilla qubits: $n = 3$
- Phase register qubits: q_2 (MSB), q_1, q_0 (LSB)
- Binary representation: $\varphi = 1/8 = 0.001_2$
- Expected outcome: integer $f = 1$, i.e. string 001, giving $\tilde{\varphi} = 1/8$ with probability 1

6.1 Step 0: Powers of T and Their Action on $|1\rangle$

We need $T^{2^0} = T$, $T^{2^1} = T^2$, and $T^{2^2} = T^4$. Computing:

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\pi/4} \end{pmatrix}, \qquad T^2 = \begin{pmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\pi/2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix} \text{ (= S gate)}, \qquad T^4 = \begin{pmatrix} 1 & 0 \\ 0 & e^{\mathrm{i}\pi} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \text{ (= Z ga}$$

Verification using phase kickback formula. For any $k \in \{0, 1, 2\}$:

$$T^{2^k}\,|1\rangle = e^{\mathrm{i}\pi 2^k/4}\,|1\rangle = e^{2\pi\mathrm{i}\cdot 2^k/8}\,|1\rangle\,,$$

so phase kickback on ancilla k will deposit the factor $e^{2\pi\mathrm{i}\cdot 2^k/8}$ onto the $|1\rangle$ branch of that qubit. Explicitly:

Qubit	Gate applied	Action on $|1\rangle$	Phase deposited
q_0 (LSB)	$T^{2^0} = T$	$e^{\mathrm{i}\pi/4}\,|1\rangle$	$e^{2\pi\mathrm{i}\cdot 1/8}$
q_1	$T^{2^1} = S$	$\mathrm{i}\,|1\rangle$	$e^{2\pi\mathrm{i}\cdot 2/8}$
q_2 (MSB)	$T^{2^2} = Z$	$-\,|1\rangle$	$e^{2\pi\mathrm{i}\cdot 4/8}$

6.2 Step 1: Initial State

$$|\Psi_0\rangle = |0\rangle_{q_2}\,|0\rangle_{q_1}\,|0\rangle_{q_0} \otimes |1\rangle = |000\rangle \otimes |1\rangle\,.$$

6.3 Step 2: Apply Hadamard to Each Ancilla

Recall $H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$. Applying H to each qubit:

$$|\Psi_1\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes \frac{|0\rangle + |1\rangle}{\sqrt{2}} \otimes |1\rangle .$$

Expanding all $2^3 = 8$ terms of the phase register:

$$|\Psi_1\rangle = \frac{1}{\sqrt{8}}\big(|000\rangle + |001\rangle + |010\rangle + |011\rangle + |100\rangle + |101\rangle + |110\rangle + |111\rangle\big) \otimes |1\rangle .$$

6.4 Step 3: Apply Controlled-T^4 (q_2 controls Z)

When $q_2 = 1$, apply Z to the eigenstate register.

$$Z|1\rangle = -|1\rangle = e^{\mathrm{i}\pi}|1\rangle = e^{2\pi\mathrm{i}\cdot 4/8}|1\rangle .$$

Phase kickback: the phase $-1 = e^{2\pi\mathrm{i}\cdot 4/8}$ appears on the $|1\rangle$ component of q_2. Only q_2 is affected; q_1 and q_0 are unchanged:

$$q_2 : \frac{|0\rangle + |1\rangle}{\sqrt{2}} \longmapsto \frac{|0\rangle + e^{2\pi\mathrm{i}\cdot 4/8}|1\rangle}{\sqrt{2}} = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

After this gate:

$$|\Psi_{2a}\rangle = \underbrace{\frac{|0\rangle - |1\rangle}{\sqrt{2}}}_{q_2} \otimes \underbrace{\frac{|0\rangle + |1\rangle}{\sqrt{2}}}_{q_1} \otimes \underbrace{\frac{|0\rangle + |1\rangle}{\sqrt{2}}}_{q_0} \otimes |1\rangle .$$

6.5 Step 4: Apply Controlled-T^2 (q_1 controls S)

When $q_1 = 1$, apply S to the eigenstate register.

$$S|1\rangle = \mathrm{i}|1\rangle = e^{\mathrm{i}\pi/2}|1\rangle = e^{2\pi\mathrm{i}\cdot 2/8}|1\rangle .$$

Phase kickback on q_1:

$$q_1 : \frac{|0\rangle + |1\rangle}{\sqrt{2}} \longmapsto \frac{|0\rangle + e^{2\pi\mathrm{i}\cdot 2/8}|1\rangle}{\sqrt{2}} = \frac{|0\rangle + \mathrm{i}|1\rangle}{\sqrt{2}}.$$

After this gate (q_2 and q_0 unchanged):

$$|\Psi_{2b}\rangle = \underbrace{\frac{|0\rangle - |1\rangle}{\sqrt{2}}}_{q_2} \otimes \underbrace{\frac{|0\rangle + \mathrm{i}|1\rangle}{\sqrt{2}}}_{q_1} \otimes \underbrace{\frac{|0\rangle + |1\rangle}{\sqrt{2}}}_{q_0} \otimes |1\rangle .$$

6.6 Step 5: Apply Controlled-T^1 (q_0 controls T)

When $q_0 = 1$, apply T to the eigenstate register.

$$T\,|1\rangle = e^{\mathrm{i}\pi/4}\,|1\rangle = e^{2\pi\mathrm{i}\cdot 1/8}\,|1\rangle\,.$$

Phase kickback on q_0:

$$q_0 : \frac{|0\rangle + |1\rangle}{\sqrt{2}} \longmapsto \frac{|0\rangle + e^{2\pi\mathrm{i}\cdot 1/8}\,|1\rangle}{\sqrt{2}} = \frac{|0\rangle + e^{\mathrm{i}\pi/4}\,|1\rangle}{\sqrt{2}}.$$

After all three controlled gates:

$$|\Psi_2\rangle = \underbrace{\frac{|0\rangle - |1\rangle}{\sqrt{2}}}_{q_2} \otimes \underbrace{\frac{|0\rangle + \mathrm{i}\,|1\rangle}{\sqrt{2}}}_{q_1} \otimes \underbrace{\frac{|0\rangle + e^{\mathrm{i}\pi/4}\,|1\rangle}{\sqrt{2}}}_{q_0} \otimes |1\rangle\,. \tag{4}$$

Expanding $|\Psi_2\rangle$ to verify the general formula. The coefficient of $|j_2 j_1 j_0\rangle$ in the phase register is:

$$\frac{1}{\sqrt{8}} \cdot (-1)^{j_2} \cdot \mathrm{i}^{j_1} \cdot e^{\mathrm{i}\pi j_0/4} = \frac{1}{\sqrt{8}} \cdot e^{\mathrm{i}\pi j_2} \cdot e^{\mathrm{i}\pi j_1/2} \cdot e^{\mathrm{i}\pi j_0/4} = \frac{1}{\sqrt{8}} e^{\mathrm{i}\pi(4j_2+2j_1+j_0)/4} = \frac{1}{\sqrt{8}} e^{2\pi\mathrm{i}j/8},$$

where $j = 4j_2 + 2j_1 + j_0$. This confirms Eq. (2) with $\varphi = 1/8$:

$$\begin{aligned} |\Psi_2\rangle &= \frac{1}{\sqrt{8}} \sum_{j=0}^{7} e^{2\pi\mathrm{i}j/8}\,|j\rangle \otimes |1\rangle \\ &= \frac{1}{\sqrt{8}}\big(|0\rangle + e^{\mathrm{i}\pi/4}\,|1\rangle + e^{\mathrm{i}\pi/2}\,|2\rangle + e^{3\mathrm{i}\pi/4}\,|3\rangle + e^{\mathrm{i}\pi}\,|4\rangle + e^{5\mathrm{i}\pi/4}\,|5\rangle + e^{3\mathrm{i}\pi/2}\,|6\rangle + e^{7\mathrm{i}\pi/4}\,|7\rangle\big) \otimes |1\rangle\,. \end{aligned} \tag{5}$$

Written with explicit binary strings (phase register only):

$$\begin{aligned} |\Psi_2\rangle\big|_{\text{phase}} = \frac{1}{\sqrt{8}}\big(&|000\rangle + e^{\mathrm{i}\pi/4}\,|001\rangle + \mathrm{i}\,|010\rangle + \mathrm{i}e^{\mathrm{i}\pi/4}\,|011\rangle \\ &- |100\rangle - e^{\mathrm{i}\pi/4}\,|101\rangle - \mathrm{i}\,|110\rangle - \mathrm{i}e^{\mathrm{i}\pi/4}\,|111\rangle\big). \end{aligned}$$

6.7 Step 6: Apply the 3-Qubit Inverse QFT

We apply QFT^{-1} to the phase register state $|\Phi\rangle = \frac{1}{\sqrt{8}} \sum_{j=0}^{7} e^{2\pi\mathrm{i}j/8}\,|j\rangle$.

Using the QFT^{-1} formula from Eq. (3) with $\varphi = 1/8$:

$$|\Psi_3\rangle\big|_{\text{phase}} = \frac{1}{8} \sum_{k=0}^{7} \left[\sum_{j=0}^{7} e^{2\pi\mathrm{i}j(1/8-k/8)} \right] |k\rangle = \frac{1}{8} \sum_{k=0}^{7} \left[\sum_{j=0}^{7} e^{2\pi\mathrm{i}j(1-k)/8} \right] |k\rangle\,.$$

Let $\omega = e^{2\pi\mathrm{i}/8}$ be the primitive 8th root of unity. The inner sum is $S_k = \sum_{j=0}^{7} \omega^{j(1-k)}$. Using the geometric series formula:

k	$1-k \pmod 8$	$S_k = \sum_{j=0}^{7} \omega^{j(1-k)}$
0	1	$\frac{1-\omega^8}{1-\omega} = 0 \quad (\omega^8 = 1,\ \omega \neq 1)$
1	0	$\sum_{j=0}^{7} 1 = 8$
2	$-1 \equiv 7$	$\frac{1-\omega^{-8}}{1-\omega^{-1}} = 0$
$3, \dots, 7$	nonzero	0

Only $k = 1$ gives a non-zero sum, so:

$$|\Psi_3\rangle\big|_{\text{phase}} = \frac{1}{8} \cdot 8 \cdot |1\rangle = |1\rangle = |001\rangle\,.$$

Therefore:

$$\boxed{|\Psi_3\rangle = |001\rangle \otimes |1\rangle\,.}$$

6.8 Step 7: Measurement and Reading the Result

Measuring the phase register yields the string **001** with probability $|\langle 001|\Psi_3\rangle|^2 = 1$.

Reading off $\tilde{\varphi}$: Interpret the measured integer $k = 1$ as a binary fraction:

$$\tilde{\varphi} = \frac{k}{2^n} = \frac{1}{2^3} = \frac{1}{8} = 0.001_2.$$

The eigenstate register is left in the state $|1\rangle$, undisturbed throughout.

Result

$$\boxed{\tilde{\varphi} = \frac{1}{8} = \varphi \checkmark}$$

QPE recovers the phase **exactly** because $\varphi = 1/8$ is representable as a 3-bit binary fraction. The measurement outcome $|001\rangle$ occurs with probability 1. The eigenstate $|1\rangle$ is returned intact.

6.9 Complete State Table

Interpretation of the table: After Step 2c, each qubit individually encodes one bit of the phase $\varphi = 0.001_2$ as a relative phase: q_2 has phase $e^{2\pi i \cdot 4/8}$ (bit 0 of φ in binary, from the MSB position), q_1 has phase $e^{2\pi i \cdot 2/8}$ (middle bit), and q_0 has phase $e^{2\pi i \cdot 1/8}$ (LSB position). The QFT^{-1} collectively decodes these three single-qubit phases into the joint basis state $|001\rangle$.

6.10 Connection to the QFT Product Formula

We can also verify the result by checking that $|\Psi_2\rangle\,|_{\text{phase}}$ is exactly $\text{QFT}\,|001\rangle$, which would directly imply $\text{QFT}^{-1}\,|\Psi_2\rangle\,|_{\text{phase}} = |001\rangle$.

Using the QFT definition with $j = 1$ (i.e. the state $|001\rangle = |j_2 = 0, j_1 = 0, j_0 = 1\rangle$):

$$\text{QFT}\,|1\rangle = \frac{1}{\sqrt{8}} \sum_{k=0}^{7} e^{2\pi i \cdot 1 \cdot k/8}\,|k\rangle = \frac{1}{\sqrt{8}} \sum_{k=0}^{7} e^{2\pi i k/8}\,|k\rangle\,,$$

Table 2: State of each qubit in the phase register at every stage of the T-gate QPE example. The eigenstate register remains $|1\rangle$ throughout.

Stage	q_2 **(MSB, controls** $T^4 = Z$**)**	q_1 **(controls** $T^2 = S$**)**	q_0 **(LSB, controls** T**)**
0. Init	$\|0\rangle$	$\|0\rangle$	$\|0\rangle$
1. After $H^{\otimes 3}$	$\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$
2a. After c-T^4	$\frac{\|0\rangle - \|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$
2b. After c-T^2	$\frac{\|0\rangle - \|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle + \mathrm{i}\,\|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle + \|1\rangle}{\sqrt{2}}$
2c. After c-T	$\frac{\|0\rangle - \|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle + \mathrm{i}\,\|1\rangle}{\sqrt{2}}$	$\frac{\|0\rangle + e^{\mathrm{i}\pi/4}\,\|1\rangle}{\sqrt{2}}$
3. After QFT^{-1}	$\|0\rangle$	$\|0\rangle$	$\|1\rangle$
4. Measurement	0	0	1

which matches Eq. (5) exactly. QFT can also be verified via the product formula with $j_1 = j_2 = 0, j_3 = 1$ ($n = 3$, bits indexed $j_1 j_2 j_3$ from MSB):

$$\mathrm{QFT}\,|001\rangle = \frac{1}{\sqrt{8}}\Big(|0\rangle + e^{2\pi\mathrm{i}\cdot 0.1}\,|1\rangle\Big) \otimes \Big(|0\rangle + e^{2\pi\mathrm{i}\cdot 0.01}\,|1\rangle\Big) \otimes \Big(|0\rangle + e^{2\pi\mathrm{i}\cdot 0.001}\,|1\rangle\Big),$$

where $0.1_2 = 1/2$, $0.01_2 = 1/4$, $0.001_2 = 1/8$, giving:

$$= \frac{1}{\sqrt{8}}\Big(|0\rangle + e^{\mathrm{i}\pi}\,|1\rangle\Big)\otimes\Big(|0\rangle + e^{\mathrm{i}\pi/2}\,|1\rangle\Big)\otimes\Big(|0\rangle + e^{\mathrm{i}\pi/4}\,|1\rangle\Big) = \frac{1}{\sqrt{8}}\Big(|0\rangle - |1\rangle\Big)\otimes\Big(|0\rangle + \mathrm{i}\,|1\rangle\Big)\otimes\Big(|0\rangle + e^{\mathrm{i}\pi/4}\,|1\rangle\Big),$$

which matches Eq. (4) exactly. This confirms the theoretical prediction: Stage 2 produces $\mathrm{QFT}\,|f\rangle$ and Stage 3 undoes it via the QFT^{-1}, recovering $|f\rangle = |001\rangle$.

7 Non-Exact Phase: A Second Worked Example

We now analyse what happens when φ is *not* exactly representable. Take U to be some unitary with $\varphi = 1/3$ and $n = 3$ ancilla qubits.

7.1 After the Controlled Gates

The general formula gives:

$$|\Psi_2\rangle\,\big|_{\text{phase}} = \frac{1}{\sqrt{8}} \sum_{j=0}^{7} e^{2\pi i j/3} |j\rangle .$$

Note that $\varphi \cdot 2^3 = 8/3 \approx 2.667$, so the nearest representable values are $k = 3$ (distance $3/8 - 1/3 = 1/24 \approx 0.042$) and $k = 2$ (distance $1/3 - 2/8 = 1/12 \approx 0.083$).

7.2 Probability Computation

For each $k \in \{0, \ldots, 7\}$, $P(k) = |\alpha_k|^2$ where:

$$\alpha_k = \frac{1}{8} \sum_{j=0}^{7} e^{2\pi i j(1/3 - k/8)}.$$

Using the closed form (Appendix A):

$$|\alpha_k|^2 = \frac{\sin^2\big(8\pi(1/3 - k/8)\big)}{64 \sin^2\big(\pi(1/3 - k/8)\big)}.$$

This is computed numerically for all k:

k	$\tilde{\varphi} = k/8$	Error $\lvert\tilde{\varphi} - 1/3\rvert$	$P(k)$
0	0.000	0.3333	≈ 0.028
1	0.125	0.2083	≈ 0.015
2	0.250	0.0833	≈ 0.188
3	0.375	0.0417	≈ 0.601
4	0.500	0.1667	≈ 0.051
5	0.625	0.2917	≈ 0.020
6	0.750	0.4167	≈ 0.063
7	0.875	0.4583	≈ 0.034
		Total	≈ 1.000

The dominant outcome is $k = 3$ (probability $\approx 60\%$), giving $\tilde{\varphi} = 3/8 = 0.375$. The error is $|0.375 - 1/3| = 1/24 \approx 0.042$, which is less than the resolution $1/2^3 = 0.125$. The second most likely outcome $k = 2$ gives $\tilde{\varphi} = 0.25$ with error $1/12 \approx 0.083$, also within the resolution guarantee. Increasing n to 4 would halve the resolution to $1/16 = 0.0625$ and sharpen the peak at $k \approx \lfloor 16/3 \rceil = 5$ ($5/16 = 0.3125$, error ≈ 0.021).

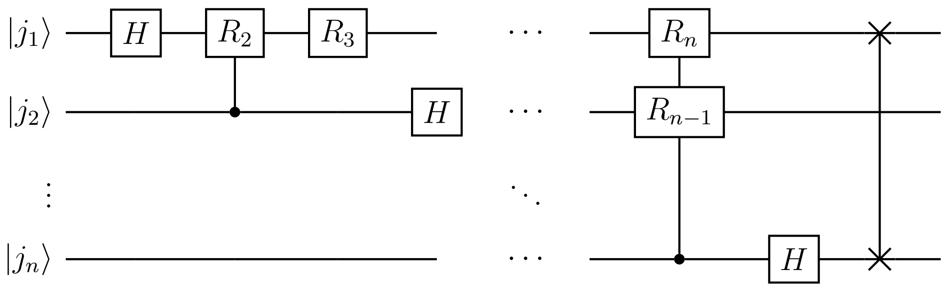
8 The Inverse QFT Circuit

The QFT is itself a quantum algorithm. Since QPE uses the QFT^{-1}, we must understand its structure.

8.1 The QFT Circuit Gates

The n-qubit QFT uses:

- n Hadamard gates H.
- $\frac{n(n-1)}{2}$ controlled phase gates $R_m = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi \mathrm{i}/2^m} \end{pmatrix}$ for $m = 2, 3, \ldots, n$, where R_m adds phase $e^{2\pi \mathrm{i}/2^m}$ when the control qubit is $|1\rangle$.
- $\lfloor n/2 \rfloor$ SWAP gates (to reverse bit order at the output).



The QFT^{-1} is the *time-reversed circuit with all phase angles negated*: replace each R_m by $R_m^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & e^{-2\pi \mathrm{i}/2^m} \end{pmatrix}$ and run the gates in reverse order.

8.2 Why Does the QFT Circuit Work?

The QFT maps $|j_1 j_2 \cdots j_n\rangle$ to the product form

$$\frac{1}{\sqrt{2^n}} \Big(|0\rangle + e^{2\pi \mathrm{i}\, 0.j_n} |1\rangle \Big) \otimes \Big(|0\rangle + e^{2\pi \mathrm{i}\, 0.j_{n-1} j_n} |1\rangle \Big) \otimes \cdots \otimes \Big(|0\rangle + e^{2\pi \mathrm{i}\, 0.j_1 j_2 \cdots j_n} |1\rangle \Big).$$

The circuit builds this factor by factor:

1. **H on qubit 1:** $|j_1\rangle \mapsto \frac{1}{\sqrt{2}}(|0\rangle + (-1)^{j_1} |1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi \mathrm{i} \cdot 0.j_1} |1\rangle)$.
2. **Controlled-R_2 from qubit 2:** multiplies the $|1\rangle$ component by $e^{2\pi \mathrm{i} j_2/4}$ when qubit 2 is in $|1\rangle$, giving $e^{2\pi \mathrm{i}\, 0.j_1 j_2}$ in the exponent.
3. **Successive controlled-R_m from qubit m:** each appends one more bit to the binary fraction in the exponent.
4. **Repeat for each qubit**, then **SWAP** to reverse bit order.

8.3 QFT Verification: 3-Qubit QFT of $|001\rangle$

We verify that $\mathrm{QFT}\,|001\rangle$ matches $|\Psi_2\rangle\,|_{\text{phase}}$ from our example (Section 6.10), confirming that the QFT^{-1} recovers $|001\rangle$ exactly.

For $|001\rangle$: $j_1 = 0$, $j_2 = 0$, $j_3 = 1$, so $j = 1$. By definition:

$$\mathrm{QFT}\,|001\rangle = \frac{1}{\sqrt{8}} \sum_{k=0}^{7} e^{2\pi \mathrm{i} \cdot k/8} \,|k\rangle\,,$$

which matches $|\Psi_2\rangle\,|_{\text{phase}}$ in Eq. (5). ✓

9 QPE With a General (Mixed Eigenstate) Input

So far we assumed $|\psi\rangle$ is a *pure eigenstate*. In practice, one often can only prepare a state with *partial overlap* with the desired eigenstate. We now analyse this case.

9.1 Superposition of Eigenstates

Let $\{U|u_j\rangle = e^{2\pi i \varphi_j}|u_j\rangle\}_{j=0,1,\dots}$ be the full set of eigenvalue/eigenstate pairs of U. Suppose the eigenstate register is initialised to:

$$|\chi\rangle = \sum_j c_j |u_j\rangle, \quad \sum_j |c_j|^2 = 1.$$

By linearity of the QPE circuit:

$$|0\rangle^{\otimes n} \otimes |\chi\rangle = \sum_j c_j \, |0\rangle^{\otimes n} \otimes |u_j\rangle \xrightarrow{\text{QPE}} \sum_j c_j \, |\tilde{\varphi}_j\rangle \otimes |u_j\rangle,$$

where $|\tilde{\varphi}_j\rangle$ is the phase-register state for eigenphase φ_j (a peak centred at $\lfloor \varphi_j \cdot 2^n \rceil$).

9.2 Measurement Outcomes

Measuring the phase register:

- With probability $|c_j|^2$, we measure (approximately) φ_j and the eigenstate register collapses to $|u_j\rangle$.
- Each run of QPE *samples* one eigenphase, weighted by $|c_j|^2$.
- If one eigenstate dominates ($|c_0|^2 \approx 1$), QPE reliably returns φ_0 with high probability.

This mechanism is used in the **Variational Quantum Eigensolver (VQE)** and quantum chemistry applications: one prepares a trial state $|\chi\rangle$ with large overlap $|c_0|^2$ with the ground state $|u_0\rangle$ of the Hamiltonian H, then runs QPE on $U = e^{-iHt}$ to estimate the ground-state energy $E_0 = 2\pi\varphi_0/t$. The better the trial state, the more reliably QPE returns the ground-state energy.

10 Applications of Quantum Phase Estimation

QPE is arguably the most important quantum subroutine. Here we survey its most significant applications.

10.1 Shor's Algorithm for Integer Factorisation

Shor's algorithm [2] factors an integer N in polynomial time. The key step: find the *period* r of the function $f(x) = a^x \bmod N$. This is done by constructing the unitary

$$U_a |x\rangle = |ax \bmod N\rangle$$

and noting that the eigenstates of U_a are $|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} \left|a^k \bmod N\right\rangle$ with eigenvalues $e^{2\pi i s/r}$ for $s = 0, 1, \ldots, r-1$. QPE extracts these eigenvalues; from s/r (as a fraction), r is recovered via the continued-fraction algorithm. Factoring then follows from r using classical number theory. Overall complexity: $\mathcal{O}((\log N)^3)$ vs. $\mathcal{O}\big(\exp\big((\log N)^{1/3}(\log\log N)^{2/3}\big)\big)$ for the best classical algorithm.

10.2 Quantum Simulation

Given a Hamiltonian H, the time-evolution operator $U = e^{-iHt}$ is unitary with eigenvalues $e^{-iE_k t}$, where E_k are energy eigenvalues. QPE on U extracts the energy spectrum, enabling:

- Molecular ground-state energies (quantum chemistry / drug discovery).
- Condensed-matter band structures and phase transitions.
- Lattice gauge theory simulations.

10.3 HHL Algorithm for Linear Systems

The Harrow–Hassidim–Lloyd (HHL) algorithm [4] solves $A\boldsymbol{x} = \boldsymbol{b}$ for a sparse Hermitian A in time $\mathcal{O}(\log N \cdot \kappa^2 s^2/\epsilon)$ (where κ is the condition number and s the sparsity), vs. $\mathcal{O}(N\kappa s)$ classically. The core subroutine is QPE applied to e^{iAt}, which extracts the eigenvalues λ_j of A; a rotation then applies $1/\lambda_j$, implementing A^{-1} in superposition.

10.4 Quantum Counting

QPE applied to the Grover iterator G extracts θ where $\sin^2\theta = M/N$ (M = number of solutions, N = search space size). From θ, M is computed exactly. This gives a quadratic speedup in counting solutions compared to classical sampling.

Table 3: Summary of QPE applications. In each case, the phase encodes the quantity of interest, and extracting the phase solves the underlying problem.

Application	**Unitary U**	**Phase encodes**
Shor's factoring	$U_a \|x\rangle = \|ax \bmod N\rangle$	Period r of $a^x \bmod N$
Quantum chemistry	e^{-iHt}	Energy eigenvalue E_k
HHL linear systems	e^{iAt}	Eigenvalue λ_j of A
Quantum counting	Grover iterator G	Number of solutions M

11 Practical Considerations for QPE

11.1 Implementing Controlled-U^{2^k} in Practice

The main cost of QPE comes from the controlled-U^{2^k} gates. For large k, 2^k is exponentially large. Two strategies:

1. **Repeated squaring.** Compute U^{2^k} by squaring k times: $U^2 = U \cdot U$, $U^4 = (U^2)^2$, etc. Cost: $\mathcal{O}(n)$ matrix multiplications (or circuit compositions) to get all powers.

2. **Problem-specific structure.** For quantum simulation, $e^{-\mathrm{i}H \cdot 2^k t} = e^{-\mathrm{i}H(2^k t)}$. This can be implemented using Trotter–Suzuki decompositions applied to the longer time $2^k t$.

11.2 Ancilla Qubit Count vs. Precision

The trade-off is fundamental:

- Each additional ancilla qubit *doubles* the precision (resolution $1/2^n$).
- For chemistry applications, "chemical accuracy" requires $|\tilde{\varphi} - \varphi| \lesssim 10^{-3}$, demanding at least $\lceil \log_2(10^3) \rceil = 10$ ancilla qubits.
- Each ancilla qubit adds one layer of controlled gates, increasing circuit depth linearly in n.

11.3 Noise and Error Mitigation on NISQ Devices

On current noisy intermediate-scale quantum (NISQ) devices, every gate has error rate $\epsilon_g \sim 10^{-3}$ to 10^{-2}. A QPE circuit with $\mathcal{O}(n^2)$ gates accumulates total error $\sim n^2 \epsilon_g$, which grows quickly with precision. Mitigation strategies include:

- **Iterative QPE (IQPE):** Estimate one bit of φ at a time using a single ancilla qubit, iterating n times. Reduces the maximum circuit depth from $\mathcal{O}(n^2)$ to $\mathcal{O}(n)$ at the cost of n sequential measurements.
- **Hadamard test:** Estimate $\mathrm{Re}\langle\psi| U^k |\psi\rangle$ and $\mathrm{Im}\langle\psi| U^k |\psi\rangle$ separately using single-ancilla circuits, then use classical post-processing to recover φ.
- **Error-mitigation techniques:** Zero-noise extrapolation, probabilistic error cancellation.

11.4 Iterative QPE (IQPE)

IQPE estimates $\varphi = 0.\omega_1\omega_2 \cdots \omega_n$ one bit at a time, using only *one ancilla qubit*. The idea: the LSB of $2^{n-1}\varphi$ (in binary) is revealed by measuring the ancilla after applying controlled-$U^{2^{n-1}}$. Classical feedback then corrects for the already-extracted bits before estimating the next.

Algorithm 1 Iterative QPE (IQPE)

Require: Oracle for controlled-U; eigenstate $|\psi\rangle$; desired bits n
Ensure: n-bit estimate $\tilde{\varphi} = 0.\omega_1\omega_2\cdots\omega_n$

```
Φ ← 0                                              ▷ Accumulated classical feedback angle
for k = n, n−1, ..., 1 do
    Prepare ancilla: |+⟩ ← H|0⟩                     ▷ Create superposition
    Apply controlled-U^(2^(k−1)) to |ψ⟩ with ancilla as control   ▷ Phase kickback
    Apply R_Z(−Φ) to ancilla                        ▷ Undo accumulated phase from prior bits
    Apply H to ancilla                              ▷ Interference to read out bit
    Measure ancilla; store result as ω_(n−k+1) ∈ {0,1}
    Φ ← (Φ + ω_(n−k+1) · π)/2                       ▷ Update feedback for next round
end for
return φ̃ = Σ_(k=1)^n ω_k / 2^k
```

12 Complete Algorithm Summary

Summary

Quantum Phase Estimation — Complete Description

Input: Black-box unitary U; (approximate) eigenstate $|\psi\rangle$ with $U|\psi\rangle = e^{2\pi i\varphi}|\psi\rangle$; desired precision n bits.

Output: $\tilde{\varphi} \in \{0, 1/2^n, \ldots, (2^n-1)/2^n\}$ such that $|\tilde{\varphi} - \varphi| \leq 1/2^n$ with probability $\geq 4/\pi^2$.

Algorithm:

1. **Initialise:** $|0\rangle^{\otimes n} \otimes |\psi\rangle$.
2. **Superposition:** Apply $H^{\otimes n}$ to the phase register.
3. **Kickback:** For $k = 0, \ldots, n-1$: apply controlled-U^{2^k} (control: ancilla qubit k, target: eigenstate register).
4. **Decode:** Apply QFT^{-1} to the phase register.
5. **Measure:** Measure the phase register; output $\tilde{\varphi} = (\text{result})/2^n$.

Why each step:

- Step 2 creates superposition so phase kickback can act on all $|j\rangle$ simultaneously.
- Step 3 encodes the phase φ into the exponentials $e^{2\pi i\varphi j}$, producing $\mathrm{QFT}\,|\varphi \cdot 2^n\rangle$.
- Step 4 inverts the QFT, converting the frequency-domain state back to the computational basis state $|f\rangle$ with $f \approx \varphi \cdot 2^n$.
- Step 5 reads out f and recovers $\tilde{\varphi} = f/2^n$.

Cost:

- $n + \mathcal{O}(n^2)$ elementary gates (Hadamards + QFT^{-1}).
- $\mathcal{O}(n)$ queries to controlled-U (using repeated squaring).
- Single-shot success probability $\geq 4/\pi^2 \approx 40.5\%$.
- Success probability $\geq 1 - \delta$ using $\mathcal{O}(\log(1/\delta))$ repetitions.

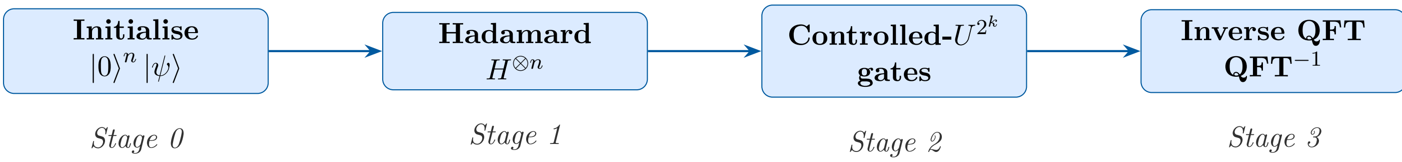


Figure 2: High-level flow of the QPE algorithm. Stage 2 encodes the phase information; Stage 3 decodes it.

13 Common Mistakes and How to Avoid Them

Mistake 1: Confusing φ with the eigenvalue. The eigenvalue is $\lambda = e^{2\pi i\varphi}$; QPE estimates $\varphi \in [0, 1)$, not λ itself. To recover the eigenvalue from the measurement result k, compute $\lambda \approx e^{2\pi ik/2^n}$.

Mistake 2: Wrong bit ordering. There are two common conventions for which ancilla qubit is MSB vs. LSB. In this lecture (following Nielsen & Chuang), qubit q_0 is the LSB and controls U^{2^0}. Some texts use the opposite convention. A circuit built with the wrong convention outputs the *bit-reversal* of the correct answer — still a valid binary string, so the error is silent and numerically plausible. Always check: the qubit controlling the *lowest* power U^1 must be read as the LSB.

Mistake 3: Forgetting the QFT bit-reversal SWAP. The standard QFT circuit produces output with bits in *reversed* order relative to the input. The SWAP gates at the end correct this. If you implement the QFT without SWAPs and use it as the forward transform, you must also implement the QFT^{-1} without SWAPs; they must be *consistent*. Mixing the conventions produces a QPE circuit that outputs the bit-reversal of φ, which is a different number.

Mistake 4: Assuming $|\psi\rangle$ is always an exact eigenstate. In practice, the input state often has only partial overlap with the target eigenstate. Section 9 shows that QPE still works, but probabilistically samples among the eigenstates with probabilities proportional to the squared overlaps $|c_j|^2$. This is not an error in the algorithm; it is the expected behaviour.

Mistake 5: Conflating circuit depth with oracle query count. QPE requires $\mathcal{O}(n)$ controlled-U oracle calls (by squaring). But if U has no special structure, each controlled-U^{2^k} may require $\mathcal{O}(2^k)$ elementary two-qubit gates, so the gate complexity can be $\mathcal{O}(2^n g)$ in the worst case. The oracle complexity and gate complexity can differ exponentially. Always state which complexity measure you are using.

14 Review Questions and Exercises

14.1 Conceptual Questions

Q1. Phase kickback. Explain in your own words why the phase $e^{2\pi i\varphi}$ is kicked back onto the *control* qubit and not the target qubit. What property of $|\psi\rangle$ is essential? What happens if the target is not an eigenstate?

Q2. Why Hadamard first? If we omit the initial Hadamard layer and apply the controlled gates to $|0\rangle^{\otimes n}$, what state do we get? Why does the QFT^{-1} then fail to give us φ?

Q3. Why powers of 2? Suppose we replace the controlled-U^{2^k} gates with controlled-U^k (powers $1, 2, 3, \dots, n$). Does the QFT^{-1} still decode the phase? Explain why not.

Q4. Precision vs. qubits. How many ancilla qubits are needed to estimate a phase to within 10^{-6} with success probability at least 99%? Show your work.

Q5. The inverse QFT. Why do we apply the *inverse* QFT rather than the forward QFT? What would the phase register state be after applying the forward QFT instead of the QFT^{-1}? (*Hint:* what is $\mathrm{QFT} \circ \mathrm{QFT}\,|f\rangle$?)

14.2 Computational Exercises

E1. Z gate. Run QPE on $U = Z$ with eigenstate $|1\rangle$ using $n = 2$ ancilla qubits. Trace all states explicitly (as in Section 6), verify the measurement outcome, and extract φ. (Answer: $\varphi = 1/2$, outcome $|10\rangle$.)

E2. S gate. Run QPE on $U = S = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix}$ with $|\psi\rangle = |1\rangle$ and $n = 3$ qubits. The eigenvalue is $\mathrm{i} = e^{\mathrm{i}\pi/2} = e^{2\pi\mathrm{i}/4}$, so $\varphi = 1/4 = 0.010_2$. Verify that the measurement outcome is $|010\rangle$ (i.e., $k = 2$, $\tilde{\varphi} = 2/8 = 1/4$).

E3. Non-exact phase. Set $\varphi = 1/3$ and $n = 3$. Compute $P(k)$ for all $k \in \{0, \dots, 7\}$ using the closed-form formula in Appendix A, and verify that $P(3) \geq 4/\pi^2$.

E4. Doubled precision. Repeat Exercise E1 with $n = 4$ qubits. What is the new resolution? Does the extra qubit change the measurement outcome? (*Hint:* $1/2 = 8/16$; the outcome should be $|1000\rangle$.)

E5. Hadamard test. Show that the circuit: (a) H on ancilla, (b) controlled-U with ancilla as control, (c) H on ancilla, (d) measure, gives $P(\text{ancilla} = 0) = \frac{1}{2}(1 + \mathrm{Re}\,\langle\psi|\,U\,|\psi\rangle)$. How is this related to the first step of QPE?

14.3 Challenging Problems

C1. Success probability lower bound. Prove rigorously that $P(b) \geq 4/\pi^2$ where $b = \lfloor \varphi \cdot 2^n \rceil$. *Hint:* Use the closed form $P(b) = \sin^2(\pi 2^n \delta)/(2^n \sin(\pi\delta))^2$ with $|\delta| \leq 1/2^{n+1}$, then bound $2^n |\sin(\pi\delta)| \leq 2^n \cdot |\pi\delta| \leq \pi/2$.

C2. Boosting success probability. Given a QPE circuit with single-shot success probability $p \geq 4/\pi^2 > 1/3$, show that repeating r times and taking the majority vote gives success probability $\geq 1 - e^{-\Omega(r)}$. (*Hint:* Apply the Chernoff bound to r i.i.d. Bernoulli trials with bias $p > 1/2$.)

C3. Shor's period finding. Verify that the states $|u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i sk/r} \left| a^k \bmod N \right\rangle$ are eigenstates of U_a with eigenvalues $e^{2\pi i s/r}$. Then show that the uniform superposition $\sum_{s=0}^{r-1} |u_s\rangle / \sqrt{r} = |1\rangle$, and explain how this means QPE on $|1\rangle$ extracts a uniformly random s/r (from which r is found via continued fractions).

C4. Iterative QPE. Implement the IQPE algorithm (Section 11.4) in a quantum circuit simulator of your choice (e.g. Qiskit, Cirq, or QuTiP). Verify it recovers $\varphi = 1/8$ exactly using a single ancilla qubit and $n = 3$ rounds. Also test it on $\varphi = 1/3$ with $n = 5$ rounds and report the estimate and error.

15 Further Reading

- **Foundational text:** Nielsen and Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2010), Chapter 5. Chapter 5.2 covers QPE and Chapter 5.3 covers applications. This is the standard reference.
- **Shor's algorithm:** P.W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," SIAM J. Comput. 26(5), 1997. Essential for seeing QPE in action.
- **HHL algorithm:** A.W. Harrow, A. Hassidim, S. Lloyd, "Quantum algorithm for linear systems of equations," PRL 103(15), 2009.
- **Quantum chemistry:** A. Aspuru-Guzik et al., "Simulated quantum computation of molecular energies," Science 309(5741), 2005. The first paper to propose QPE for quantum chemistry.
- **Iterative QPE:** A.Yu. Kitaev, "Quantum measurements and the abelian stabilizer problem," arXiv:quant-ph/9511026, 1995. The original IQPE proposal.
- **Modern review:** S. McArdle et al., "Quantum computational chemistry," Rev. Mod. Phys. 92(1), 2020. Comprehensive coverage of QPE in near-term quantum computing.
- **Qiskit tutorials:** `https://qiskit.org/learn/` — worked QPE implementations in Python with IBM Quantum hardware.
- **Quirk:** `https://algassert.com/quirk` — a browser-based quantum circuit simulator excellent for visualising QPE interactively.

16 Key Equations Cheat Sheet

Setting: U unitary; $U|\psi\rangle = e^{2\pi i\varphi}|\psi\rangle$; $\varphi \in [0,1)$; n ancilla qubits; $N = 2^n$.

Phase kickback (k-th ancilla):

$$c\text{-}U^{2^k}\,\frac{|0\rangle + |1\rangle}{\sqrt{2}}\,|\psi\rangle = \frac{|0\rangle + e^{2\pi i\varphi 2^k}\,|1\rangle}{\sqrt{2}}\,|\psi\rangle\,.$$

State after all controlled gates (Stage 2):

$$|\Psi_2\rangle = \frac{1}{\sqrt{N}}\sum_{j=0}^{N-1} e^{2\pi i\varphi j}\,|j\rangle \otimes |\psi\rangle\,.$$

This equals QFT $|f\rangle \otimes |\psi\rangle$ where $f = \varphi N$ (when exact), so QFT^{-1} recovers $|f\rangle$.

Probability amplitude after $\mathbf{QFT^{-1}}$:

$$\alpha_k = \frac{1}{N}\sum_{j=0}^{N-1} e^{2\pi i j(\varphi - k/N)}.$$

Probability of outcome k:

$$P(k) = |\alpha_k|^2 = \frac{\sin^2\!\big(\pi N(\varphi - k/N)\big)}{N^2\sin^2\!\big(\pi(\varphi - k/N)\big)}.$$

Exact case ($\varphi = f/N$, integer f): $P(f) = 1$, all others 0.
Worst-case success probability: $P(b) \geq 4/\pi^2 \approx 40.5\%$, where $b = \lfloor \varphi N \rceil$.
Resolution: $|\tilde{\varphi} - \varphi| \leq 1/N = 1/2^n$.
Gate count: $\mathcal{O}(n^2 + ng)$ total; $\mathcal{O}(n)$ queries to U.

QFT: $\mathrm{QFT}\,|j\rangle = \frac{1}{\sqrt{N}}\sum_{k=0}^{N-1} e^{2\pi i jk/N}\,|k\rangle$.
$\mathbf{QFT^{-1}}$: $\mathrm{QFT}^{-1}\,|j\rangle = \frac{1}{\sqrt{N}}\sum_{k=0}^{N-1} e^{-2\pi i jk/N}\,|k\rangle$.

Binary fraction: $\varphi = 0.b_1 b_2 \cdots b_n = \sum_{k=1}^{n} b_k/2^k$.

A Geometric Series Closed Form

For the probability amplitude computation, we repeatedly use:

$$\sum_{j=0}^{N-1} r^j = \frac{1-r^N}{1-r}, \quad r \neq 1, \quad N = 2^n.$$

With $r = e^{2\pi \mathrm{i}\alpha}$ for non-integer α, using $|1-e^{\mathrm{i}\theta}| = 2|\sin(\theta/2)|$:

$$\left|\sum_{j=0}^{N-1} e^{2\pi \mathrm{i} j\alpha}\right|^2 = \frac{|1-e^{2\pi \mathrm{i} N\alpha}|^2}{|1-e^{2\pi\mathrm{i}\alpha}|^2} = \frac{4\sin^2(\pi N\alpha)}{4\sin^2(\pi\alpha)} = \frac{\sin^2(\pi N\alpha)}{\sin^2(\pi\alpha)}.$$

When $\alpha = \varphi - k/N$ (as in the QPE probability computation), this gives:

$$P(k) = \frac{1}{N^2} \cdot \frac{\sin^2\big(\pi N(\varphi - k/N)\big)}{\sin^2\big(\pi(\varphi - k/N)\big)}.$$

B Binary Fraction Expansion

Any $\varphi \in [0,1)$ has the binary expansion $\varphi = \sum_{k=1}^{\infty} b_k 2^{-k}$ with $b_k \in \{0,1\}$. The n-bit truncation $f/2^n = \sum_{k=1}^{n} b_k 2^{-k}$ satisfies $|\varphi - f/2^n| \leq 2^{-n}$, with equality when all remaining bits $b_{n+1}, b_{n+2}, \ldots$ are 1.

Conversion examples:

φ	Exact binary	$n=3$ approx	f (integer)	Error $\lvert\varphi - f/8\rvert$
$1/2$	0.1_2	0.100_2	4	0
$1/4$	0.01_2	0.010_2	2	0
$1/8$	0.001_2	0.001_2	1	0
$3/8$	0.011_2	0.011_2	3	0
$1/3$	$0.\overline{01}_2$	0.011_2	3	$1/24 \approx 0.042$
$1/5$	$0.\overline{0011}_2$	0.001_2	1	$3/40 = 0.075$

C Verification of the T-Gate Example by Direct Matrix Computation

For completeness, we verify $\mathrm{QFT}^{-1}\,|\Psi_2\rangle\,|_{\text{phase}} = |001\rangle$ directly using the 8×8 DFT matrix.

The phase-register state before the QFT^{-1} is (from Eq. (5)):

$$|\Psi_2\rangle\,\big|_{\text{phase}} = \frac{1}{\sqrt{8}}\big[1,\, \omega,\, \omega^2,\, \omega^3,\, \omega^4,\, \omega^5,\, \omega^6,\, \omega^7\big]^T, \quad \omega = e^{2\pi\mathrm{i}/8}.$$

The vector of coefficients is $\boldsymbol{v} = \frac{1}{\sqrt{8}}[\omega^j]_{j=0}^{7}$.

The QFT^{-1} matrix has entries $(F^{-1})_{kj} = \frac{1}{\sqrt{8}} e^{-2\pi\mathrm{i}kj/8} = \frac{\omega^{-kj}}{\sqrt{8}}$. Then:

$$(F^{-1}\boldsymbol{v})_k = \sum_{j=0}^{7} \frac{\omega^{-kj}}{\sqrt{8}} \cdot \frac{\omega^j}{\sqrt{8}} = \frac{1}{8}\sum_{j=0}^{7} \omega^{j(1-k)}.$$

By the geometric series formula:

$$\frac{1}{8}\sum_{j=0}^{7}\omega^{j(1-k)} = \delta_{k,1},$$

since $\omega^8 = 1$ (so the sum equals 8 for $k = 1$ and 0 otherwise).

Therefore $F^{-1}\boldsymbol{v} = \boldsymbol{e}_1 = [0, 1, 0, 0, 0, 0, 0, 0]^T$, corresponding to the state $|001\rangle$. □

D Second Worked Example: The S Gate with $n = 2$ Ancilla Qubits

This shorter example reinforces every concept from Section 6 with a different gate and fewer qubits, so it is easy to check on paper.

Setup

- Unitary: $U = S = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix}$
- Eigenstate: $|\psi\rangle = |1\rangle$, eigenvalue $S|1\rangle = \mathrm{i}\,|1\rangle = e^{\mathrm{i}\pi/2}|1\rangle$
- True phase: $\varphi = 1/4 = 0.01_2$
- Ancilla qubits: $n = 2$, so $N = 4$ representable values: $\{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$
- Phase register: q_1 (MSB), q_0 (LSB)
- Expected outcome: integer $f = 1$, string $|01\rangle$, giving $\tilde{\varphi} = 1/4$

Powers of S:

$$S = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i} \end{pmatrix}, \qquad S^2 = \begin{pmatrix} 1 & 0 \\ 0 & \mathrm{i}^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = Z.$$

Action on the eigenstate:

$$S^{2^0}|1\rangle = \mathrm{i}\,|1\rangle = e^{2\pi\mathrm{i}\cdot 1/4}|1\rangle\,, \qquad S^{2^1}|1\rangle = Z\,|1\rangle = -\,|1\rangle = e^{2\pi\mathrm{i}\cdot 2/4}|1\rangle\,.$$

Step 1 — Initial state: $|\Psi_0\rangle = |00\rangle \otimes |1\rangle$.

Step 2 — After $H^{\otimes 2}$:

$$|\Psi_1\rangle = \frac{1}{2}\big(|00\rangle + |01\rangle + |10\rangle + |11\rangle\big) \otimes |1\rangle = \frac{1}{2}\sum_{j=0}^{3}|j\rangle \otimes |1\rangle\,.$$

Step 3 — Apply controlled-$S^2 = Z$ (q_1 controls):

Phase $e^{2\pi\mathrm{i}\cdot 2/4} = e^{\mathrm{i}\pi} = -1$ kicks back onto q_1:

$$q_1 : \frac{|0\rangle + |1\rangle}{\sqrt{2}} \longmapsto \frac{|0\rangle + e^{\mathrm{i}\pi}|1\rangle}{\sqrt{2}} = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$

Step 4 — Apply controlled-$S^1 = S$ (q_0 controls):

Phase $e^{2\pi \mathrm{i} \cdot 1/4} = \mathrm{i}$ kicks back onto q_0:

$$q_0 : \ \frac{|0\rangle + |1\rangle}{\sqrt{2}} \longmapsto \frac{|0\rangle + \mathrm{i}\,|1\rangle}{\sqrt{2}}.$$

State after all controlled gates:

$$|\Psi_2\rangle = \underbrace{\frac{|0\rangle - |1\rangle}{\sqrt{2}}}_{q_1} \otimes \underbrace{\frac{|0\rangle + \mathrm{i}\,|1\rangle}{\sqrt{2}}}_{q_0} \otimes |1\rangle\,.$$

Expanding (4 terms):

$$|\Psi_2\rangle\Big|_{\text{phase}} = \frac{1}{2}\big(|00\rangle + \mathrm{i}\,|01\rangle - |10\rangle - \mathrm{i}\,|11\rangle\big) = \frac{1}{2}\big(1\cdot|0\rangle + \mathrm{i}^1\,|1\rangle + \mathrm{i}^2\,|2\rangle + \mathrm{i}^3\,|3\rangle\big) = \frac{1}{2}\sum_{j=0}^{3} e^{2\pi \mathrm{i} j/4}\,|j\rangle\,.$$

This confirms the general formula with $\varphi = 1/4$. ✓

Step 5 — Apply QFT^{-1} to the 2-qubit phase register:

The 2-qubit QFT^{-1} has matrix entries $(F_4^{-1})_{kj} = \frac{1}{2}e^{-2\pi \mathrm{i} kj/4}$. Compute each amplitude:

$$\alpha_k = \frac{1}{4}\sum_{j=0}^{3} e^{2\pi \mathrm{i} j(1/4 - k/4)} = \frac{1}{4}\sum_{j=0}^{3} \mathrm{i}^{j(1-k)}.$$

k	$\sum_{j=0}^{3} \mathrm{i}^{j(1-k)}$	Value
0	$\mathrm{i}^0 + \mathrm{i}^1 + \mathrm{i}^2 + \mathrm{i}^3 = 1 + \mathrm{i} - 1 - \mathrm{i}$	$= 0$
1	$\mathrm{i}^0 + \mathrm{i}^0 + \mathrm{i}^0 + \mathrm{i}^0 = 1 + 1 + 1 + 1$	$= 4$
2	$\mathrm{i}^0 + \mathrm{i}^{-1} + \mathrm{i}^{-2} + \mathrm{i}^{-3} = 1 - \mathrm{i} - 1 + \mathrm{i}$	$= 0$
3	$\mathrm{i}^0 + \mathrm{i}^{-2} + \mathrm{i}^{-4} + \mathrm{i}^{-6} = 1 - 1 + 1 - 1$	$= 0$

Therefore $|\Psi_3\rangle = \frac{1}{4}\cdot 4 \cdot |1\rangle \otimes |1\rangle = |01\rangle \otimes |1\rangle$.

Measurement: Outcome $|01\rangle$, i.e. $k = 1$, $\tilde{\varphi} = 1/4$.

$$\boxed{\tilde{\varphi} = \frac{1}{4} = \varphi \ ✓}$$

E Shor's Algorithm: Period-Finding via QPE, Step by Step

We walk through the smallest non-trivial factoring example — $N = 15$, $a = 7$ — at full algebraic detail.

E.1 Problem Statement

Factor $N = 15$. Choose $a = 7$ with $\gcd(7, 15) = 1$. We seek the *order* (period) r of 7 modulo 15: the smallest positive integer r such that $7^r \equiv 1 \pmod{15}$.

Computing directly: $7^1 = 7$, $7^2 = 49 \equiv 4$, $7^3 \equiv 28 \equiv 13$, $7^4 \equiv 91 \equiv 1 \pmod{15}$. So $r = 4$.

E.2 Constructing the Unitary U_a

Define the unitary U_7 acting on a 4-qubit register (representing $\{0, 1, \ldots, 15\}$) by:

$$U_7 |x\rangle = |7x \bmod 15\rangle \quad \text{for } x \in \{1, \ldots, 14\}, \qquad U_7 |0\rangle = |0\rangle .$$

The orbit of 1 under repeated application is: $1 \to 7 \to 4 \to 13 \to 1 \to \cdots$ (period 4).

E.3 Eigenstates and Eigenvalues of U_7

The eigenstates of U_7 (restricted to the orbit of 1) are:

$$|u_s\rangle = \frac{1}{2} \sum_{k=0}^{3} e^{-2\pi i sk/4} \left|7^k \bmod 15\right\rangle, \quad s = 0, 1, 2, 3.$$

Explicitly:

$$|u_s\rangle = \frac{1}{2}\big(|1\rangle + e^{-2\pi i s/4} |7\rangle + e^{-2\pi i s\cdot 2/4} |4\rangle + e^{-2\pi i s\cdot 3/4} |13\rangle\big).$$

One can verify $U_7 |u_s\rangle = e^{2\pi i s/4} |u_s\rangle$, so the eigenvalues are $e^{2\pi i \varphi_s}$ with $\varphi_s = s/r = s/4$:

$$\varphi_0 = 0, \quad \varphi_1 = \tfrac{1}{4}, \quad \varphi_2 = \tfrac{1}{2}, \quad \varphi_3 = \tfrac{3}{4}.$$

E.4 The QPE Input State

We cannot easily prepare $|u_s\rangle$ for a specific s. Instead, note:

$$\frac{1}{2} \sum_{s=0}^{3} |u_s\rangle = \frac{1}{4} \sum_{s=0}^{3} \sum_{k=0}^{3} e^{-2\pi i sk/4} \left|7^k \bmod 15\right\rangle = |1\rangle ,$$

since the sum over s produces a delta function at $k = 0$. So we input $|1\rangle$ to the eigenstate register — a state with equal weight $|c_s|^2 = 1/4$ on all four eigenstates.

E.5 QPE Output and Period Extraction

With $n = 4$ ancilla qubits (so $N = 16 \geq r^2 = 16$; we need $N \geq r^2$ for the continued-fraction step to succeed reliably), QPE produces the phase estimate $\tilde{\varphi}_s \approx s/4$ for each s with equal probability.

Each possible measurement outcome $k \in \{0, 1, \ldots, 15\}$ gives $\tilde{\varphi} = k/16$. The continued-fraction algorithm applied to $k/16$ finds the best rational approximation with denominator $\leq \sqrt{16} = 4$:

Measured k	$\tilde{\varphi} = k/16$	Best rational approx	Gives period
0	$0/16 = 0$	$0/1$	trivial (r not found)
4	$4/16 = 1/4$	$1/4$	$r = 4$
8	$8/16 = 1/2$	$1/2$	$r = 2$ (divisor of 4)
12	$12/16 = 3/4$	$3/4$	$r = 4$

From $r = 4$: $\gcd(7^2 - 1, 15) = \gcd(48, 15) = \gcd(3, 15) = 3$ and $\gcd(7^2 + 1, 15) = \gcd(50, 15) = 5$. So $15 = 3 \times 5$. $\square$

Why $n \geq 2\log_2 r$? The continued-fraction step needs to distinguish s/r from nearby fractions with denominator $< r$. By the theory of continued fractions, if $|k/N - s/r| < 1/(2r^2)$, then s/r is the unique best rational approximation. With $N = 2^n \geq r^2$, the resolution $1/N \leq 1/r^2 < 1/(2r^2)$ after rounding, guaranteeing the condition is met.

F Spectral Leakage and the Dirichlet Kernel

F.1 The QPE Probability as a Dirichlet Kernel

When φ is not exactly representable, the probability distribution $P(k)$ is not a perfect spike. The distribution is precisely characterised by the *Dirichlet kernel*, a classical signal-processing object.

Define the shifted variable $\delta_k = \varphi - k/N$ (where $N = 2^n$). From Appendix A:

$$P(k) = \frac{1}{N^2} \cdot \frac{\sin^2(\pi N \delta_k)}{\sin^2(\pi \delta_k)} =: \frac{1}{N^2} \mathcal{D}_N(\delta_k),$$

where $\mathcal{D}_N(\delta) = \frac{\sin^2(\pi N\delta)}{\sin^2(\pi\delta)}$ is the **squared Dirichlet kernel**. Key properties:

- $\mathcal{D}_N(0) = N^2$ (the exact case: probability 1 at $k = f$).
- $\mathcal{D}_N(m/N) = 0$ for any non-zero integer m (perfect destructive interference at all other grid points when φ is exact).
- The "main lobe" has width $\approx 2/N$ (the probability is concentrated in 2 grid spacings around the peak).
- "Side lobes" decay as $\sim 1/(\pi N \delta_k)^2$ for large $|\delta_k|$.

F.2 Why Side Lobes Appear

The side lobe structure is the quantum analogue of spectral leakage in the Fast Fourier Transform. When φ is not on the grid, the "frequency" does not fit exactly into the N-point DFT window, and energy leaks into neighbouring bins. The falloff rate $\sim 1/\delta_k^2$ is relatively slow — this is the same poor windowing as a rectangular window in classical signal processing.

F.3 Tail Probability Bound

How much probability leaks beyond distance ℓ from the peak?

$$\sum_{|k-b|>\ell} P(k) \;\leq\; \sum_{|k-b|>\ell} \frac{1}{(\pi N |\delta_k|)^2} \;\approx\; \frac{1}{\pi^2 \ell}.$$

For $\ell = 1/\epsilon - 1/2$, the tail probability is at most $\epsilon/(\pi^2/4)$. This is used to derive the precise precision theorem: to guarantee $|\tilde{\varphi} - \varphi| \leq 1/2^t$ with probability $\geq 1 - \epsilon$, use $n = t + \lceil \log_2(2 + 1/(2\epsilon)) \rceil$ qubits.

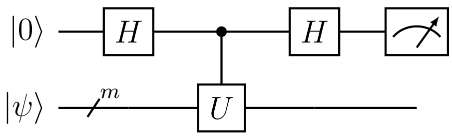
F.4 Improving the Probability: Amplitude Amplification

The 40.5% worst-case success probability can be improved without adding ancilla qubits by using *quantum amplitude amplification* (a generalisation of Grover's algorithm). Applying $\lceil \pi/(4\theta) \rceil$ amplification steps (where $\sin^2\theta = p$) boosts $p \to\approx 1$ at a cost of $\mathcal{O}(1/\sqrt{p})$ additional oracle calls. For $p = 4/\pi^2$, this gives $\mathcal{O}(1)$ additional calls.

G The Hadamard Test and Its Relation to QPE

G.1 The Single-Qubit Hadamard Test

The **Hadamard test** is the simplest quantum circuit for estimating an expectation value of a unitary U with respect to a state $|\psi\rangle$.



Analysis. The ancilla evolves as:

$$\begin{aligned}
|0\rangle|\psi\rangle &\xrightarrow{H\otimes\mathbb{1}} \frac{|0\rangle+|1\rangle}{\sqrt{2}}|\psi\rangle \\
&\xrightarrow{c\text{-}U} \frac{|0\rangle|\psi\rangle+|1\rangle U|\psi\rangle}{\sqrt{2}} \\
&\xrightarrow{H\otimes\mathbb{1}} \frac{(|0\rangle+|1\rangle)|\psi\rangle+(|0\rangle-|1\rangle)U|\psi\rangle}{2} \\
&= |0\rangle\frac{|\psi\rangle+U|\psi\rangle}{2}+|1\rangle\frac{|\psi\rangle-U|\psi\rangle}{2}.
\end{aligned}$$

The probability of measuring 0 is:

$$P(0)=\left\|\frac{|\psi\rangle+U|\psi\rangle}{2}\right\|^2=\frac{1+\mathrm{Re}\,\langle\psi|U|\psi\rangle}{2}.$$

Similarly, $P(0)$ for the circuit with an extra $S^\dagger$ gate before the final H gives $\frac{1+\mathrm{Im}\langle\psi|U|\psi\rangle}{2}$.

G.2 Connection to QPE

The Hadamard test and QPE are deeply related:

- Each step of **iterative QPE** is a Hadamard test with $U^{2^{k-1}}$.
- The full QPE circuit can be viewed as n simultaneous Hadamard tests (one per ancilla qubit), where the QFT^{-1} coherently combines their outputs.
- The Hadamard test estimates $\mathrm{Re}\,\langle\psi|U^k|\psi\rangle$, which equals $\mathrm{Re}\,e^{2\pi i\varphi k}=\cos(2\pi\varphi k)$ when $|\psi\rangle$ is an eigenstate. The QFT^{-1} then inverts the discrete Fourier series $\sum_k\cos(2\pi\varphi k)$, recovering φ.

Operational view. The n-qubit QPE circuit simultaneously estimates n Fourier coefficients $e^{2\pi i\varphi\cdot 2^k}$ for $k=0,\ldots,n-1$ via phase kickback. The QFT^{-1} then coherently combines these into a single sharp peak at φ. This is analogous to how the FFT combines n single-frequency measurements into a full spectrum.

H Fault-Tolerant QPE and Quantum Resource Estimation

H.1 Why Fault Tolerance is Necessary

Current quantum hardware has gate error rates $\epsilon_g \sim 10^{-3}$. A QPE circuit for a 50-qubit Hamiltonian (relevant for small molecules in quantum chemistry) might require $n = 20$ ancilla qubits and $\mathcal{O}(n \cdot 2^{50})$ controlled-U applications (before squaring) — leading to catastrophic error accumulation.

Fault-tolerant quantum computing (FTQC) uses quantum error-correcting codes to suppress logical error rates to arbitrarily small values. The leading paradigm is the **surface code**.

H.2 The Surface Code and Magic State Distillation

A surface code of *code distance* d encodes 1 logical qubit into $\mathcal{O}(d^2)$ physical qubits and suppresses the logical error rate to $\epsilon_L \approx (10\,\epsilon_g)^{\lceil d/2 \rceil}$. For $\epsilon_g = 10^{-3}$ and $d = 7$: $\epsilon_L \approx 10^{-11}$.

Clifford gates (including H, S, CNOT) are transversal in the surface code. The non-Clifford T gate (our running example!) requires *magic state distillation*: prepare many noisy $|T\rangle$ states and distil one high-quality one via a distillation circuit. This dominates the resource cost of fault-tolerant QPE.

H.3 Logical Qubit and T-Gate Requirements for Quantum Chemistry QPE

For a molecule with η electrons and orbital basis of size M:

Resource	**Requirement**	**Notes**
Logical qubits (data)	$\mathcal{O}(M)$	Represent molecular orbitals
Ancilla qubits (QPE)	$n \approx 20$–50	Chemical accuracy $\approx 10^{-3}$ Ha
T-gate count	$\mathcal{O}(M^5/\epsilon)$	From Trotter simulation of $e^{-\mathrm{i}Ht}$
Physical qubits (surface code)	$\sim 10^6$–10^9	For large molecules
Runtime	Hours to days	At MHz physical gate clock

The $\mathcal{O}(M^5)$ T-gate count (Babbush et al., 2019 [8]) can be reduced to $\mathcal{O}(M^{4+o(1)})$ using qubitization techniques (Berry et al., 2019 [9]).

H.4 Qubitization: A Superior Approach to Hamiltonian Simulation

Instead of implementing $U = e^{-\mathrm{i}Ht}$ via Trotter steps, the **qubitization** approach directly encodes $H/\|H\|_1$ into a unitary walk operator W (using a technique called *block encoding*) such that:

$$W\,|0\rangle\,|v\rangle = \sin(\theta)\,|0\rangle\,|H/\|H\|_1 v\rangle + \cos(\theta)\,|1\rangle\,|\perp\rangle\,,$$

where the eigenvalues of $H/\|H\|_1$ are in $[-1, 1]$ and can be written as $\cos(\theta)$. QPE on W extracts θ, from which the energy is recovered as $E = \|H\|_1 \cos(\theta)$. This gives a *quadratic improvement* in the cost of Hamiltonian simulation: Trotter costs $\mathcal{O}(M^5/\epsilon)$ T-gates; qubitization costs $\mathcal{O}(M^5/\sqrt{\epsilon})$ or better.

I QPE Variants: A Comparative Overview

Many variants of QPE have been developed for specific use cases. This section surveys the most important ones.

Table 4: Comparison of QPE variants. Here n = desired precision bits, ϵ = failure probability, g = gate count per U query, m = eigenstate register size.

Variant	**Ancilla qubits**	**Circuit depth**	**Best for**
Standard QPE [1]	$n + m$	$\mathcal{O}(n^2 + ng)$	Fault-tolerant hardware
Iterative QPE (IQPE) [3]	$1 + m$	$\mathcal{O}(ng)$ per bit	Low-qubit devices
Textbook Kitaev QPE [3]	$\mathcal{O}(\log(1/\epsilon))$	$\mathcal{O}(1/\epsilon)$	Rigorous guarantees
Hadamard test (classical post-processing)	$1 + m$	$\mathcal{O}(2^k)$ per estimate	NISQ, many repetitions
Robust QPE (RQPE) [10]	$1 + m$	$\mathcal{O}(1/\epsilon)$	Adversarial noise
Quantum phase estimation with randomisation [11]	$\mathcal{O}(1) + m$	$\mathcal{O}(\text{poly}(1/\epsilon))$	Reduced ancilla count
Qubitization + QPE [8]	$n + \mathcal{O}(M)$	$\mathcal{O}(nM)$	Quantum chemistry
Bayesian QPE [12]	$1 + m$	adaptively chosen	Classical inference-heavy

I.1 Bayesian QPE

Bayesian QPE treats φ as a random variable with a prior distribution $p_0(\varphi)$. At each step, the experimenter adaptively chooses the power k (not necessarily a power of 2!), applies the Hadamard test, and updates the posterior using Bayes' rule:

$$p_{t+1}(\varphi) \propto P(\text{outcome}_t \mid \varphi, k_t) \cdot p_t(\varphi).$$

Because the likelihood function $P(\text{outcome} = 0 \mid \varphi, k) = \frac{1}{2}(1 + \cos(2\pi\varphi k))$ is smooth, the posterior sharpens rapidly: after $\mathcal{O}(n)$ adaptive experiments the posterior has standard deviation $\sim 2^{-n}$, matching the standard QPE precision at the same oracle cost.

The advantage: Bayesian QPE is more robust to decoherence, can use non-power-of-2 queries (allowing optimal experimental design), and provides uncertainty quantification directly.

I.2 Robust QPE

Robust QPE (Kimmel et al., 2015 [10]) is designed to work even when the controlled-U gate has coherent errors. Instead of assuming U is applied exactly, it tolerates a worst-case gate error ϵ_g per layer and still recovers φ to precision $1/2^n$ with high probability, provided $\epsilon_g \lesssim 1/n$.

J Quantum Chemistry Case Study: Ground State Energy of H_2

J.1 Physical Setup

The hydrogen molecule H_2 (two protons, two electrons) has a Hamiltonian H that, in the STO-3G minimal basis (4 spin-orbitals), can be mapped to a 2-qubit operator via the Jordan–Wigner transform (after accounting for the frozen core approximation):

$$H = h_{00}Z_0 + h_{11}Z_1 + h_{01}Z_0Z_1 + g_{0011}(X_0X_1 + Y_0Y_1),$$

where X_k, Y_k, Z_k are Pauli operators on qubit k, and the coefficients h_{ij}, g_{0011} depend on the internuclear distance R. For $R = 0.735\,\text{Å}$ (equilibrium bond length):

$$h_{00} \approx -0.4488, \quad h_{11} \approx -0.4488, \quad h_{01} \approx 0.1810, \quad g_{0011} \approx 0.1810.$$

(All values in Hartree.)

J.2 Constructing the Time-Evolution Operator

The time-evolution operator for time t is $U = e^{-\mathrm{i}Ht}$. For QPE we need controlled-U^{2^k}, i.e. controlled-$e^{-\mathrm{i}Ht2^k}$.

Using the first-order Trotter decomposition:

$$e^{-\mathrm{i}Ht} \approx e^{-\mathrm{i}h_{00}Z_0t} \cdot e^{-\mathrm{i}h_{11}Z_1t} \cdot e^{-\mathrm{i}h_{01}Z_0Z_1t} \cdot e^{-\mathrm{i}g_{0011}(X_0X_1+Y_0Y_1)t}.$$

Each factor is a standard rotation gate. For example, $e^{-\mathrm{i}\theta Z} = R_Z(2\theta)$ and $e^{-\mathrm{i}\theta(XX+YY)} = e^{-\mathrm{i}\theta XX} \cdot e^{-\mathrm{i}\theta YY}$, each decomposable into at most 6 CNOT gates and single-qubit rotations.

J.3 QPE Circuit for H_2

With $n = 8$ ancilla qubits (providing precision $\sim 4 \times 10^{-3}$ Ha, within chemical accuracy), the QPE circuit for H_2 requires:

- 8 Hadamard gates (on ancilla).
- 8 controlled-U^{2^k} operations for $k = 0, \ldots, 7$; each controlled-U^{2^k} is implemented as 2^k Trotter steps, costing $\mathcal{O}(2^k)$ CNOT gates (without squaring) or $\mathcal{O}(k)$ (with squaring).
- One 8-qubit QFT^{-1} ($\mathcal{O}(8^2) = 64$ gates).
- 8 measurements.

J.4 Expected Measurement Outcome

The ground-state energy of H_2 at equilibrium is $E_0 \approx -1.1373$ Ha. For time $t = 1$ (atomic units), the corresponding phase is:

$$\varphi_0 = \frac{E_0 t}{2\pi} + \frac{1}{2} = \frac{-1.1373}{2\pi} + 0.5 \approx 0.5 - 0.181 = 0.319,$$

where the shift by $1/2$ is applied to keep $\varphi_0 \in [0, 1)$ when $E_0 < 0$. (In practice one calibrates t so that $\varphi_0 \approx 1/4$ for maximum precision.)

With $n = 8$ qubits, $\varphi_0 \cdot 256 \approx 81.7$, so the most likely measurement outcome is $k = 82$, giving:

$$\tilde{\varphi} = \frac{82}{256} \approx 0.3203, \quad \tilde{E}_0 = 2\pi(\tilde{\varphi} - 0.5) \cdot \frac{1}{t} \approx 2\pi \cdot (-0.180) \approx -1.131 \text{ Ha}.$$

The error $|\tilde{E}_0 - E_0| \approx 0.006$ Ha is just outside chemical accuracy; using $n = 10$ would achieve $\lesssim 10^{-3}$ Ha.

Why quantum chemistry is hard classically. The H_2 example above seems trivial — only 2 qubits for the electrons! However, for a molecule with η electrons in M orbitals, the Hamiltonian acts on a $\binom{M}{\eta}$-dimensional space. For iron-sulfur clusters (relevant in nitrogen fixation), $M \sim 50$ and $\eta \sim 30$, giving Hilbert space dimension $\sim 10^{14}$ — completely infeasible classically but tractable with a fault-tolerant quantum computer using $\mathcal{O}(M)$ logical qubits.

K Extensions: Multi-Phase and Continuous-Variable QPE

K.1 Multi-Phase Estimation

Suppose U has multiple eigenvalues $e^{2\pi i\varphi_1}, \ldots, e^{2\pi i\varphi_r}$ and we wish to estimate all of them simultaneously. Running QPE with eigenstate $|\chi\rangle = \sum_j c_j |u_j\rangle$ outputs eigenphase φ_j with probability $|c_j|^2$. To estimate all r phases:

- Repeat QPE $\mathcal{O}(r \log r/\epsilon^2)$ times and collect a histogram of measurement outcomes.
- Cluster the outcomes around the r peaks. Each cluster centroid approximates one eigenphase.

This requires knowing r a priori or using a clustering algorithm such as k-means on the measurement histogram.

K.2 Continuous-Variable QPE

In the continuous-variable (CV) formalism (photonic quantum computing), qubits are replaced by *modes* with infinite-dimensional Hilbert space. The CV analogue of QPE estimates the eigenvalue of a unitary acting on a bosonic mode using *homodyne measurement* instead of computational-basis measurement. The QFT^{-1} is replaced by a phase-space rotation. CV-QPE is used in Gaussian boson sampling and can achieve Heisenberg-limited phase estimation (standard deviation $\sim 1/N$) with N photons.

K.3 Heisenberg Limit vs. Standard Quantum Limit

Standard QPE uses N queries to U and achieves precision $\sigma_\varphi \sim 1/N$. This is the **Heisenberg limit** — the best possible with N queries. By contrast, a classical "frequentist" scheme using N independent single-query experiments achieves only $\sigma_\varphi \sim 1/\sqrt{N}$ (the **standard quantum limit**). QPE achieves a quadratic improvement by exploiting quantum coherence between the N queries.

QPE is Heisenberg-optimal. One can show (via quantum Fisher information bounds) that no quantum algorithm using N controlled-U queries can achieve $\sigma_\varphi < 1/(2\pi N)$. Standard QPE matches this bound (up to constants), making it *optimal* in the query complexity sense.

L Numerical Simulation: Python Implementation of QPE

The following Python pseudocode (using NumPy and conceptual quantum operations) implements the 3-qubit QPE for the T gate, reproducing the exact calculation of Section 6.

```
import numpy as np

# Define T gate and powers
omega = np.exp(2j * np.pi / 8)      # primitive 8th root of unity
T = np.array([[1, 0], [0, omega]])   # T gate = e^{i*pi/4} on |1>

# QPE state vector simulation (phase register: 3 qubits, 8 dimensions)
N = 8   # 2^3
psi = np.zeros(N, dtype=complex)

# After H^(x3): uniform superposition
psi = np.ones(N, dtype=complex) / np.sqrt(N)

# After controlled-T^{2^k} gates: multiply basis state |j> by e^{2pi*i*phi*j}
phi = 1/8
for j in range(N):
    psi[j] *= np.exp(2j * np.pi * phi * j)

# After inverse QFT: apply N-point inverse DFT
psi = np.fft.ifft(psi) * np.sqrt(N)  # ifft includes 1/N normalisation

# Measurement probabilities
probs = np.abs(psi)**2
print("Probabilities:", np.round(probs, 6))
# Expected output: [0, 1, 0, 0, 0, 0, 0, 0]
# i.e., P(k=1) = 1, all others 0.

# Read off phase
k_measured = np.argmax(probs)
phi_est = k_measured / N
print(f"Estimated phase: {phi_est}")   # 0.125 = 1/8
```

Note that the NumPy `ifft` applies the normalisation $1/N$ while the quantum QFT^{-1} applies $1/\sqrt{N}$. The factor `np.sqrt(N)` corrects for this.

For a non-exact phase such as $\varphi = 1/3$, replace `phi = 1/8` with `phi = 1/3` and observe a peaked but spread probability distribution, matching the analysis of Section 7.

M Historical Notes: The Development of QPE

1993 — Bernstein and Vazirani. The first quantum algorithm to exploit interference for phase estimation was implicit in the Bernstein–Vazirani algorithm, which extracts a hidden bit-string from a linear function.

1994 — Shor. Peter Shor's factoring algorithm [2] was the first explicit use of what we now call QPE. He used the quantum FFT (discovered independently around the same time) to extract the period r from the eigenvalues of the modular exponentiation unitary. This paper single-handedly launched the field of quantum algorithms.

1995 — Kitaev. Alexei Kitaev formalised the QPE primitive in full generality [3], introducing the abelian stabilizer problem and the iterative variant (IQPE) using a single ancilla qubit. He also proved the precision and complexity guarantees that are now standard.

1996 — Abrams and Lloyd. Abrams and Lloyd proposed applying QPE to the time-evolution operator $e^{-\mathrm{i}Ht}$ to compute eigenvalues of quantum Hamiltonians [6], founding quantum simulation as a field.

1999 — Mosca and Ekert. Mosca and Ekert gave a clean pedagogical treatment of QPE in the context of the hidden subgroup problem, clarifying the connection to the QFT.

2005 — Aspuru-Guzik et al. The first concrete proposal to use QPE for quantum chemistry [5], estimating molecular ground-state energies — the application that continues to drive quantum hardware development.

2009 — HHL algorithm. Harrow, Hassidim, and Lloyd used QPE as a subroutine in the first exponentially-fast quantum linear-systems algorithm [4].

2018–present — Qubitization and block encoding. Berry, Babbush, and collaborators developed qubitization [8, 9], reducing the T-gate cost of QPE for chemistry from $\mathcal{O}(M^5/\epsilon)$ to $\mathcal{O}(M^{4+o(1)}/\sqrt{\epsilon})$, making fault-tolerant QPE for 100-electron molecules plausible with $\sim 10^6$ physical qubits.

N Glossary of Key Terms

Ancilla qubit An auxiliary qubit used as a control or workspace. In QPE, the n ancilla qubits form the phase register.

Binary fraction
A number of the form $0.b_1b_2\cdots b_n = \sum_{k=1}^{n} b_k 2^{-k}$. The output of QPE is a binary fraction approximating φ.

Block encoding
A technique that embeds a non-unitary matrix A as the top-left block of a larger unitary: $U = \begin{pmatrix} A/\|A\| & \cdots \\ \cdots & \cdots \end{pmatrix}$. Used in qubitization-based QPE.

Chemical accuracy
An error threshold of ≈ 1 kcal/mol $\approx 1.6 \times 10^{-3}$ Ha in energy calculations. This is the target precision for quantum chemistry applications of QPE.

Controlled-U gate
A gate that applies U to the target register if and only if the control qubit is $|1\rangle$.

Dirichlet kernel
The function $\mathcal{D}_N(\delta) = \sin^2(\pi N\delta)/\sin^2(\pi\delta)$ that characterises the QPE probability distribution.

Eigenphase The phase $\varphi \in [0, 1)$ such that the eigenvalue of the unitary U is $e^{2\pi i\varphi}$.

Eigenstate A state $|\psi\rangle$ satisfying $U|\psi\rangle = \lambda|\psi\rangle$ for some scalar λ.

Heisenberg limit
The optimal phase estimation precision of $\sigma_\varphi \sim 1/N$ achievable with N queries.

Inverse QFT (IQFT)
The adjoint of the Quantum Fourier Transform; the decoder step of QPE.

Magic state distillation
A fault-tolerant procedure for producing high-fidelity T-gate resource states from many noisy copies.

Phase kickback
The mechanism by which the phase $e^{2\pi i\varphi}$ migrates from the eigenstate register to the control qubit when the eigenstate undergoes a controlled unitary.

Phase register
The n-qubit ancilla register in QPE that stores the estimate of φ.

Quantum Fourier Transform (QFT)
The unitary $\mathrm{QFT}\,|j\rangle = \frac{1}{\sqrt{N}}\sum_k e^{2\pi i jk/N}|k\rangle$, the quantum analogue of the discrete Fourier transform.

Qubitization A technique for Hamiltonian simulation that achieves near-optimal query complexity for QPE in quantum chemistry.

Resolution The smallest difference $1/2^n$ between representable phase values with n ancilla qubits.

Spectral leakage Probability appearing at measurement outcomes far from the true phase, analogous to spectral leakage in classical FFT with finite windows.

Surface code A topological quantum error-correcting code with threshold $\sim 1\%$, the leading candidate for fault-tolerant quantum computation.

Trotter decomposition An approximation $e^{-\mathrm{i}(A+B)t} \approx e^{-\mathrm{i}At} \cdot e^{-\mathrm{i}Bt} + \mathcal{O}(t^2)$ used to simulate Hamiltonian time evolution on a quantum computer.

Unitary operator A linear operator U satisfying $U^\dagger U = \mathbb{1}$. All eigenvalues lie on the complex unit circle.

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